

Longitudinal Modeling of Cognitive Decline in Older Adults Using GLM, GEE, and GLMM: An Analysis of the OASIS-2 Cohort

AUTHOR

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1 Introduction

Alzheimer's disease (AD) is the most common cause of dementia worldwide and remains one of the most pressing public health challenges of the 21st century. Global estimates consistently show that the number of individuals with dementia will rise substantially as populations age, placing increasing demands on health systems, caregivers, and societies. International reports by Alzheimer's Disease International and the World Health Organization highlight the accelerating global burden of dementia and emphasize the importance of early identification and intervention ([1], [2]). Additional epidemiological analyses further demonstrate sharp projected increases in dementia prevalence across diverse regions, with recent summaries underscoring the significant clinical and economic implications of this trend ([3], [4]).

Accurate and early cognitive assessment plays a central role in diagnosing Alzheimer's disease, evaluating disease progression, and identifying individuals at elevated risk of cognitive decline. The Mini-Mental State Examination (MMSE) is one of the most widely used tools for assessing global cognitive status in both clinical practice and research. Originally developed as a brief, standardized screening test for cognitive impairment [5], the MMSE has been extensively validated in psychometric and methodological reviews [6]. A comprehensive Cochrane review confirms its practical diagnostic accuracy in community and primary care settings, particularly among older adults without prior cognitive evaluation [7]. Its short administration time, standardized scoring, and widespread clinical familiarity have established the MMSE as a central component of AD assessment frameworks.

A critical aspect of using the MMSE in both clinical and epidemiologic research is identifying a meaningful cutoff for cognitive impairment. A well-established threshold is a total score below 24, which corresponds to at least mild impairment and indicates elevated risk for functional decline. Validation studies show that **MMSE < 24** demonstrates high specificity for probable Alzheimer's disease, even across diverse demographic groups. In highly educated samples, the standard cutoff retains excellent specificity and acceptable sensitivity [8]. Although more recent calibration studies have proposed refined thresholds for distinguishing different impairment levels, **MMSE < 24 remains a widely used and clinically intuitive definition of poor cognition** [9]. At the same time, population-based norms reveal that MMSE performance varies systematically by age and education, highlighting the need to adjust for demographic covariates when evaluating cognitive status or decline [10].

Understanding cognitive decline requires longitudinal assessment. A growing body of evidence links demographic characteristics to differences in cognitive functioning and trajectories. Education is strongly associated with baseline cognitive performance, often interpreted through cognitive reserve theory; however, research suggests that education primarily influences initial cognitive level rather than the rate of decline [11]. Sex-specific patterns have also been documented, with

educational gradients in cognitive impairment risk varying by gender and age [12]. Additionally, life-course factors, including parental education, contribute to later-life cognitive functioning [13]. These findings underscore the importance of modeling demographic and clinical influences, including age, sex, and education, when examining cognitive outcomes.

Given this context, the present study investigates two key research questions: (a) How are baseline age, education, and sex associated with the probability of poor cognition (defined as $MMSE < 24$), and how does this probability change over time? (b) How do baseline age, education, sex, and dementia group relate to the rate of cognitive decline, as measured by longitudinal MMSE scores? By modeling cognitive impairment as a dynamic process and examining how demographic and clinical factors shape both the likelihood and progression of poor cognition, this study aims to clarify the determinants of cognitive decline and inform strategies for early risk identification and dementia-focused interventions.

To address these research aims, we utilize data from the Open Access Series of Imaging Studies (OASIS), a publicly available neuroimaging dataset designed to support research on brain aging and Alzheimer’s disease. The longitudinal structure of the OASIS dataset provides a unique opportunity to examine cognitive trajectories across a wide age range and across dementia statuses. This enables our study to investigate how baseline demographic and clinical factors, including age, sex, education, relate to both the probability and progression of poor cognition as reflected in longitudinal MMSE performance.

Methodologically, we employ three complementary modeling approaches to address our two research questions: a generalized linear model (GLM) to estimate baseline associations with the probability of poor cognition ($MMSE < 24$) and the change of MMSE ($\Delta MMSE$), and both generalized linear mixed models (GLMM) and generalized estimating equations (GEE) to analyze longitudinal changes in MMSE outcomes and the evolution of poor-cognition risk over time.

2 Methods

2.1 Dataset description

We analyzed data from the OASIS-2 longitudinal MRI aging cohort, which includes between one and four clinical visits per participant. At each visit, demographic characteristics, clinical dementia ratings (CDR), cognitive assessments (MMSE), and MRI-derived brain measures were recorded. The analytic dataset used in this project was restricted to individuals with valid MMSE measurements at baseline and at least one follow-up visit for longitudinal analysis.

OASIS provides richly detailed cross-sectional and longitudinal magnetic resonance imaging (MRI) data collected through collaborations among the Washington University Alzheimer’s Disease Research Center, the Neuroinformatics Research Group at Washington University, the Biomedical Informatics Research Network, and the Howard Hughes Medical Institute. The longitudinal dataset includes 142 adults aged 60 to 96, each with two or more MRI sessions separated by at least one year, totaling 373 imaging sessions. Within this cohort, 72 participants were nondemented across all visits, 56 were demented at baseline, and 14 transitioned from nondemented to demented during follow-up.

Table 1: Baseline Descriptive Statistics

Descriptive statistics by education level (Low / Medium / High)

Variable	Overall	Low	Medium	High
N	128	42	28	58
age	75.26 (7.74)	76.07 (7.36)	75.79 (8.54)	74.41 (7.65)
mmse	27.44 (3.08)	26.60 (3.60)	27.18 (3.12)	28.17 (2.47)
sex = F	74 (57.8%)	23 (54.8%)	19 (67.9%)	32 (55.2%)
sex = M	54 (42.2%)	19 (45.2%)	9 (32.1%)	26 (44.8%)

At baseline (visit = 1), the OASIS dataset included 142 participants across three groups (**High Education**, **Medium Education** and **Low Education**). Descriptive statistics for age, education, MMSE, and sex distribution at baseline are summarized in [Table 1](#).

2.1.1 Research Problem 1

We defined **poor cognition** as $MMSE < 24$ at the final observed visit, a commonly used threshold for cognitive impairment.

$$Y_i = \mathbb{I}(MMSE_{i,\text{final}} < 24)$$

The binary outcome (poor vs. non-poor cognition, based on MMSE) was modeled using three complementary approaches:

- **Logistic generalized linear model (GLM)**
 - Cross-sectional analysis using baseline predictors to estimate the odds of poor cognition at the final assessment.
- **Logistic generalized estimating equations (GEE)**
 - Population-averaged model for repeated MMSE-based outcomes, accounting for within-subject correlation across visits.
- **Logistic generalized linear mixed model (GLMM)**
 - Subject-specific model with random effects, providing individual-level trajectories of the probability of poor cognition over time.

2.1.2 Research Problem 2

The methods we used for this problem are:

- GLM (linear): overall change from baseline to last visit ($\Delta MMSE$)
- GEE (Gaussian, identity link): population-average MMSE trajectory over time
- GLMM (linear mixed model): subject-specific MMSE trajectories over time with random intercepts and slopes

We defined **Change in MMSE** as our final outcome in **GLM** Model. For **GEE** and **GLMM** model, we decided to use **Raw MMSE Score** as our final outcome.

2.2 Data Preprocessing

All analyses were conducted using a cleaned version of the OASIS-2 dataset. Participants missing baseline MMSE values were excluded to ensure consistent cognitive assessment across models. For

the GLM analyses, we further restricted the sample to individuals with at least two clinical visits so that baseline-to-final change could be defined. Poor cognition was operationalized as $\text{MMSE} < 24$ at the last available visit for each subject. Continuous predictors (age) were inspected for missingness and extreme values, and factor variables (sex and education group) were cleaned and converted into categorical formats. The design matrix was evaluated for multicollinearity using variance inflation factors (VIF), and no meaningful collinearity was detected. No imputation was performed, and all models were fit using complete-case data. Importantly, the same preprocessed dataset was used to construct outcomes for GLM, GEE, and GLMM, allowing for consistent comparison across cross-sectional, marginal, and subject-specific modeling frameworks.

2.3 Methods Description

2.3.1 Generalized Linear Model (GLM)

A cross-sectional logistic generalized linear model (GLM) was used to estimate the association between baseline predictors and the probability of poor cognition at the final assessment ($\text{MMSE} < 24$). The logistic GLM is **well-suited** for this **binary outcome** because it models the log-odds of impairment as a linear function of predictors, provides interpretable effect estimates in the form of odds ratios, and ensures that predicted probabilities remain between 0 and 1. This model summarizes how demographic and clinical characteristics at baseline, specifically age, education, sex, contribute to the probability of cognitive impairment in a single time point analysis. The GLM also serves as a foundational model that establishes baseline association patterns before incorporating longitudinal dependence in repeated-measures frameworks such as GEE and GLMM, offering a computationally efficient and statistically robust first step for identifying key predictors of poor cognition.

The model takes the following form:

$$\text{logit} [P(Y_i = 1)] = \beta_0 + \beta_1 \text{Age}_i + \beta_2 \text{Educ}_i + \beta_3 \text{Sex}_i.$$

The following assumptions and diagnostic checks were performed to evaluate the adequacy of the logistic GLM:

- **Correct link function**

Verified by assessing whether the logit link appropriately captured the relationship between predictors and the outcome.

- **Independence of observations**

Reasonable in this cross-sectional setting, with one observation per participant.

- **Absence of influential outliers**

Examined through standardized residuals, leverage values, and Cook's distance.

- **Multicollinearity**

Assessed using variance inflation factors (VIF) for all covariates.

- **Goodness of fit**

Evaluated using the Hosmer–Lemeshow test and associated calibration diagnostics.

- **Model discrimination**

Quantified using receiver operating characteristic (ROC) curves and the area under the curve

(AUC).

- **Calibration**

Evaluated by comparing predicted probabilities with observed event rates across decile-based prediction bins.

2.3.2 GEE

2.3.2.1 Model Specification

In our initial GLM analyses we used a main-effects model to describe the association between cognition and baseline covariates, treating observations as independent. When moving to the GEE framework, our scientific question shifted to population-average trajectories of cognition over repeated visits. Exploratory plots of MMSE by education level suggested that the mean profiles over visit were not parallel across education groups, indicating possible effect modification of time by education. To capture these differing slopes, we expanded the GEE mean model to include a visit $\times$ education interaction, while retaining the same core covariates (age, sex, and education). Thus, the GEE mean model is an extension of the GLM that allows the effect of time to vary by education level, which is only identifiable in the longitudinal setting.

We analyzed repeated Mini-Mental State Examination (MMSE) measurements across multiple study visits. Education was categorized as Low (≤ 12 years), Medium (13 – 15 years), and High (≥ 16 years). Sex and baseline age were included as adjustment covariates. Two research questions were addressed:

- RQ1: How does education level influence poor cognition ($\text{MMSE} < 24$), adjusting for age and sex?
- RQ2: How does education level influence the continuous MMSE score over time, adjusting for age and sex?

For RQ1, we fit a population-averaged logistic GEE with a binomial family and logit link:

$$\text{logit}[P(Y_{it} = 1)] = \beta_0 + \beta_1 \text{Visit}_{it} + \beta_2 \text{Educ}_i + \beta_3 (\text{Visit}_{it} \times \text{Educ}_i) + \beta_4 \text{Age}_i + \beta_5 \text{Sex}_i.$$

For RQ2, we modeled MMSE score as a continuous outcome using Gaussian GEE with an identity link:

$$E(\text{MMSE}_{it}) = \beta_0 + \beta_1 \text{Visit}_{it} + \beta_2 \text{Educ}_i + \beta_3 (\text{Visit}_{it} \times \text{Educ}_i) + \beta_4 \text{Age}_i + \beta_5 \text{Sex}_i.$$

2.3.2.2 Choosing the Working Correlation Structure

We then compared Independence, Exchangeable, and AR-1 structures using the Quasi-likelihood Information Criterion (QIC):

Model Fit Comparison	
QIC Values by Correlation Structure For RQ1	
corstr	QIC
independence	277.3027
exchangeable	275.9959

Model Fit Comparison

QIC Values by Correlation Structure For RQ1

corstr	QIC
ar1	277.9506

Model Fit Comparison

QIC Values by Correlation Structure For RQ2

corstr	QIC
independence	4358.758
exchangeable	4502.593
ar1	4618.946

As shown in **?@tbl-gee-qic**, for RQ1, the exchangeable structure had the lowest QIC and aligned well with the correlogram. And for RQ2 (see **?@tbl-gee-qic-rq2**), independence had the numerically lowest QIC, but empirical correlations demonstrated strong within-subject dependence. Therefore, we selected the Exchangeable structure for both RQ1 and RQ2 to provide a model consistent with observed correlation patterns.

2.3.3 GLMM

To address the research questions regarding the relationship between Education and Cognitive Function over time, we fit two mixed-effects models. Both models adjusted for Baseline Age (scaled) and Sex, and included random intercepts to account for within-subject correlation.

For RQ1 (Binary Outcome: Poor Cognition), a logistic Generalized Linear Mixed Model (GLMM) was fit. The model specification included interaction terms between Visit and Education Level to test if the probability of poor cognition changes differently over time based on education:

$$\text{logit}(P(Y_{ij} = 1)) = \beta_0 + \beta_1 \text{Visit}_{ij} + \beta_2 \text{Educ}_{\text{Med}} + \beta_3 \text{Educ}_{\text{High}} + \beta_4 (\text{Visit} \times \text{Educ}_{\text{Med}}) + \beta_5 (\text{Visit} \times \text{Educ}_{\text{High}}) + \beta_6 \text{Age}_i + \beta_7 \text{Sex}_i + b_{0i}$$

where $b_{0i} \sim N(0, \sigma_b^2)$.

For RQ2 (Continuous Outcome: MMSE Score), a Linear Mixed Model (LMM) was fit to examine trajectories of cognitive decline. Similar to RQ1, this model included interactions between Visit and Education:

$$\text{MMSE}_{ij} = \beta_0 + \beta_1 \text{Visit}_{ij} + \beta_2 \text{Educ}_{\text{Med}} + \beta_3 \text{Educ}_{\text{High}} + \beta_4 (\text{Visit} \times \text{Educ}_{\text{Med}}) + \beta_5 (\text{Visit} \times \text{Educ}_{\text{High}}) + \beta_6 \text{Age}_i + \beta_7 \text{Sex}_i + b_{0i} + \epsilon_{ij}$$

where the random effects (b_{0i}, b_{1i}) are assumed to be normally distributed.

Random effects:

- Random intercept (binary outcome)
- Random intercept + slope (continuous outcome)

Variance components were extracted, and intraclass correlation (ICC) was computed. The Zeger attenuation formula was used to compare conditional and marginal effects.

3 Results

3.1 Generalized Linear Model (GLM)

3.1.1 Research Question 1 (Binary Outcome): Poor Cognition Probability

3.1.1.1 Model Summary

Table 2: Point-biserial correlation between MMSE and CDR Score

estimate	statistic	p.value	conf.low	conf.high	method
-0.6983218	-10.95113	<1e-07	-0.7776202	-0.5971545	Pearson's product-moment correlation

Table 3: Point-biserial correlation between MMSE and Demented Status

estimate	statistic	p.value	conf.low	conf.high	method
-0.649699	-9.593442	<1e-07	-0.7398203	-0.5366746	Pearson's product-moment correlation

Before constructing the final model, we first investigated the correlation between our outcome (MMSE score) and potential predictors. Clinical Dementia Rating (CDR) score and dementia status were both strongly correlated with MMSE in our data (see [Table 2](#), [Table 3](#)), and previous work has shown that MMSE, CDR, and clinical dementia diagnosis largely reflect overlapping constructs of global cognitive impairment. Consequently, including CDR or dementia status as predictors of MMSE would be conceptually redundant and introduce circularity, so these variables were not considered as candidate predictors in the multivariable models.

Table 4: GLM Model Summary

term	estimate	std.error	statistic	p.value
(Intercept)	2.163	2.853	0.758	0.448
age_baseline	0.000	0.032	0.014	0.989
educ_baseline	-0.278	0.091	-3.071	0.002
sex_baselineM	0.462	0.479	0.965	0.335

As shown in [Table 4](#), we fit a logistic GLM to evaluate whether baseline demographic and clinical characteristics were associated with poor cognition (MMSE < 24) at the subject's final recorded visit. Predictors included baseline age, education, and sex. By checking p-value, only Education is significant since they have a smaller p value. Overall, the logistic GLM identified baseline education as significant predictors of poor cognition at follow-up. Higher education was protective, while greater baseline cognitive impairment strongly increased the likelihood of future poor cognition.

Table 5: Odds Ratio table of GLM model

Predictor	Odds Ratio (95% CI)
(Intercept)	8.701 (0.035, 2703.168)
age_baseline	1.000 (0.939, 1.065)
educ_baseline	0.757 (0.626, 0.896)
sex_baselineM	1.588 (0.618, 4.112)

Moreover, we also summarized the odds ratios (ORs) and 95% confidence intervals (CIs) from the logistic GLM (See [Table 5](#)), which modeled the association between baseline predictors and the likelihood of poor cognition at the subject’s final visit. Each additional year of education reduced the odds of poor cognition by approximately 25%, ($OR = 0.75$; $95\%CI : 0.62\text{--}0.89$). Baseline age and sex were not significant predictors after adjustment. These results suggest that education is an important protective factor, whereas age and sex alone do not strongly predict poor cognitive outcome when restricted to final-visit data.

3.1.1.2 Model Performance

To evaluate overall model performance, we assessed the classification accuracy by comparing predicted probabilities against the observed outcomes, providing a summary measure of how well the model distinguished individuals with and without poor cognition.

Metric	Value
Accuracy	0.828

Table 6: GLM Model Metrics Summary

From [Table 6](#), the model correctly classified most cases ($\approx 83\%$), yielding strong performance for identifying cases.

Area under the curve: 0.6841

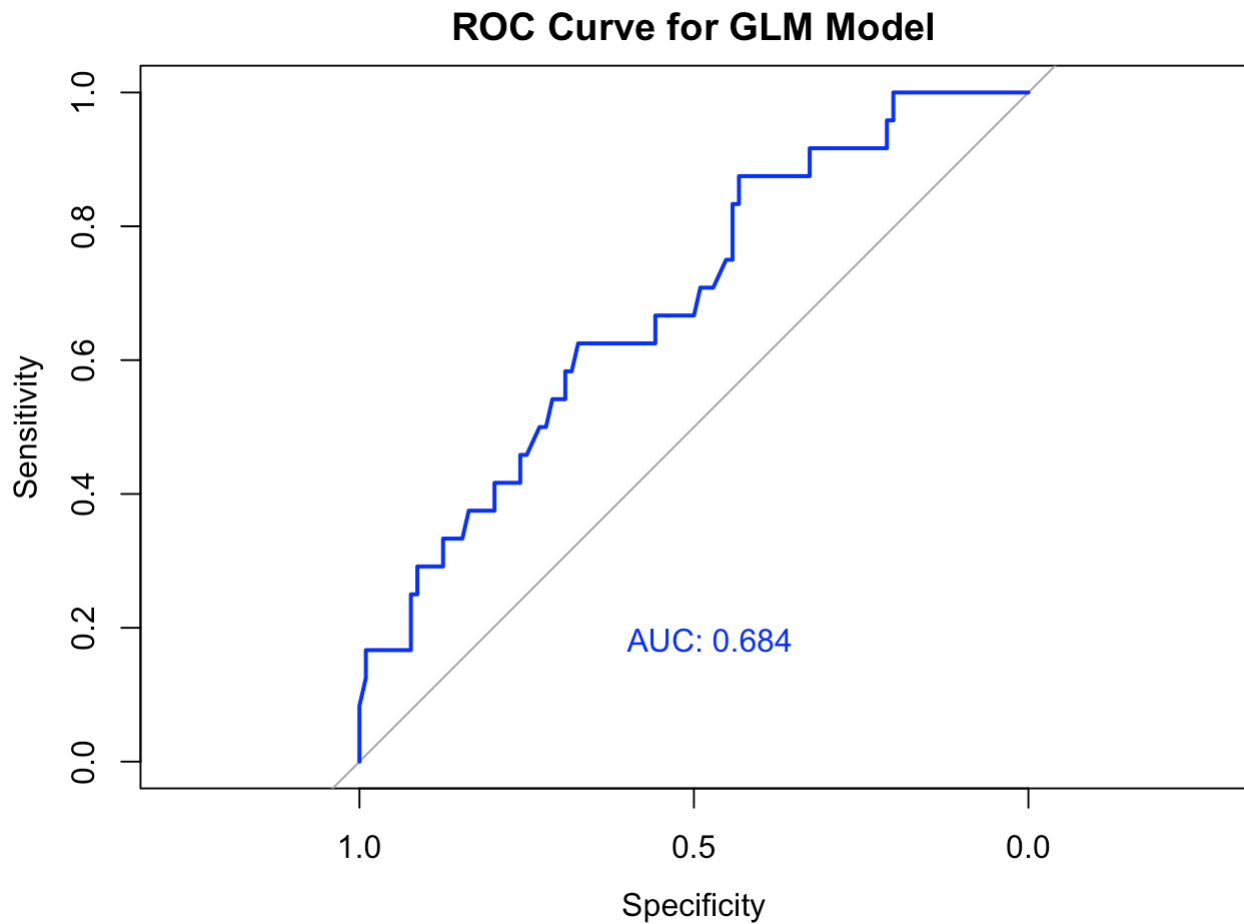


Figure 1: ROC Curve for GLM Model

The ROC curve (See [Figure 1](#)) for the logistic GLM demonstrates strong discriminative ability, with the model performing substantially better than chance. The curve rises steeply and remains well above the 45° reference line, suggesting an area under the curve (AUC) 0.684. This indicates that the model effectively separates subjects with poor cognition from those without impairment. Although the default classification threshold of 0.5 yields moderate sensitivity due to class imbalance, the ROC curve shows that alternative thresholds would substantially improve the detection of cognitively impaired individuals without sacrificing excessive specificity.

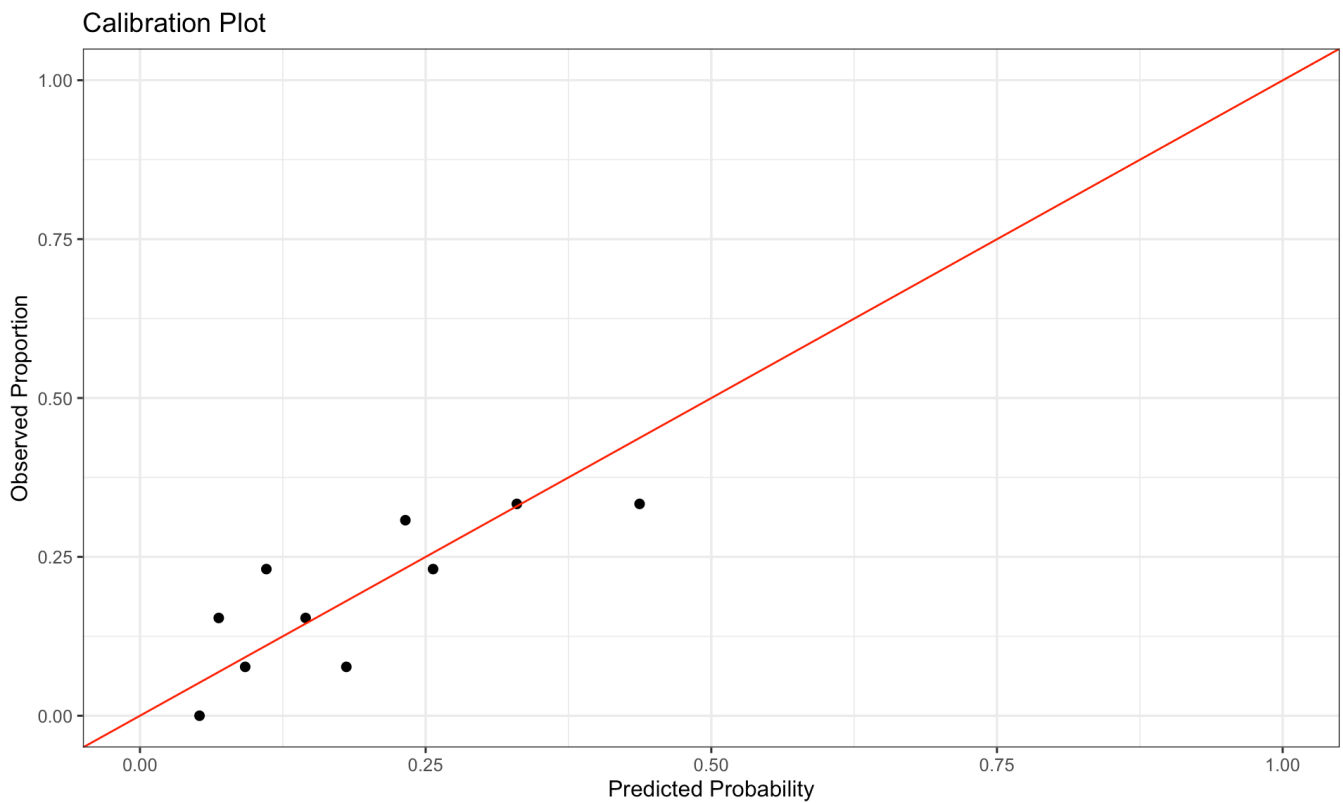


Figure 2: Calibration Plot for GLM

Table 7: Calibration Table (Decile-Based)

bin	mean_pred	obs_rate
1	0.052	0.000
2	0.069	0.154
3	0.092	0.077
4	0.111	0.231
5	0.145	0.154
6	0.181	0.077
7	0.232	0.308
8	0.257	0.231
9	0.330	0.333
10	0.437	0.333

As shown in [Table 7](#) and [Figure 2](#), the calibration plot compares predicted probabilities to observed rates of poor cognition across decile bins. Most points lie close to the 45-degree identity line, suggesting that the logistic GLM provides well-calibrated probability estimates.

Overall, The model is well calibrated for low to moderate risk levels and shows modest underestimation at the highest predicted risks.

3.1.1.3 Model diagnostic

3.1.1.3.1 Deviance Residuals

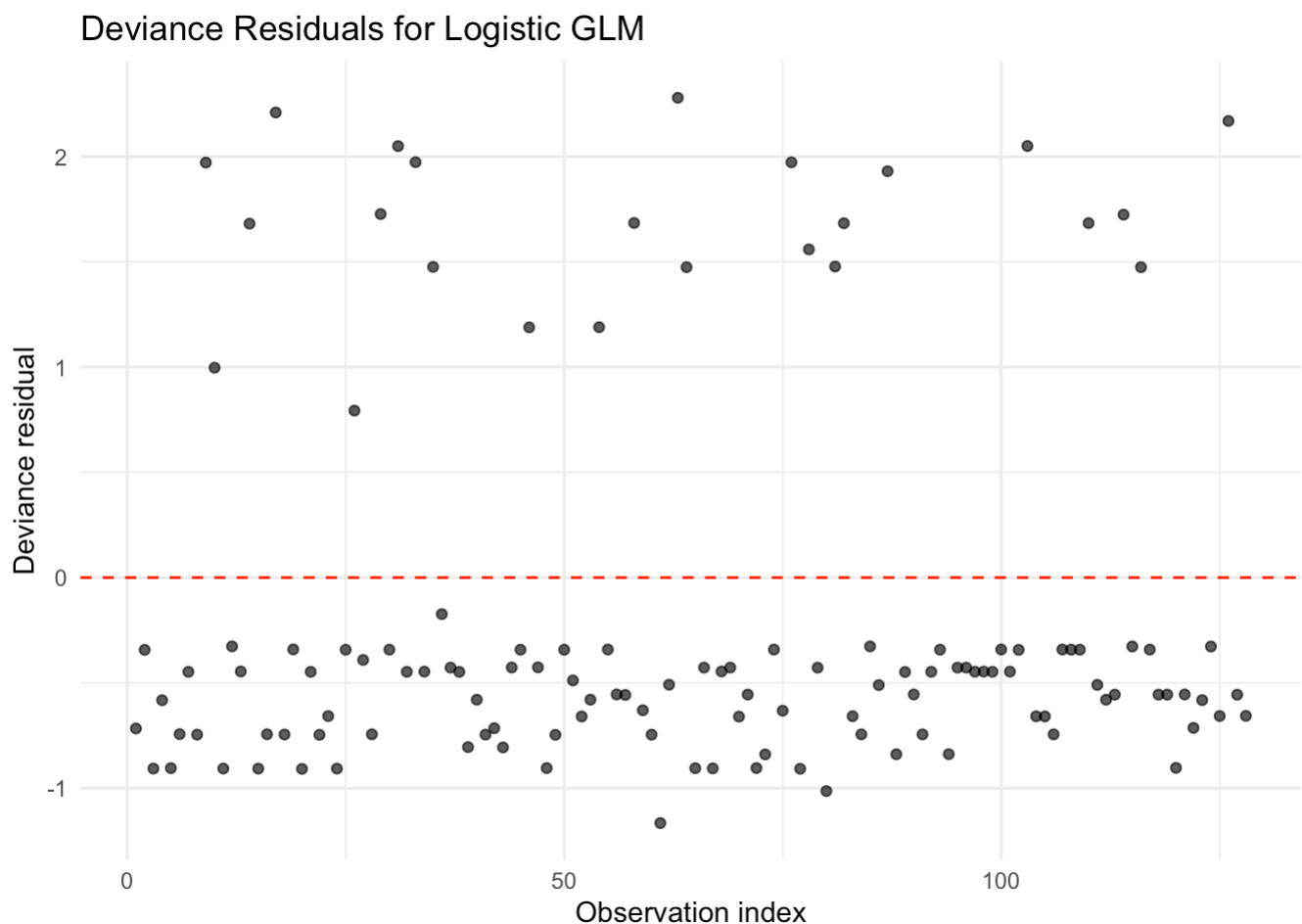


Figure 3: Deviance Residuals Plot for Logistic GLM

From [Figure 3](#), the deviance residual plot shows that most residuals fall within -1 and $+1$, suggesting that the logistic GLM provides an adequate fit to the majority of individuals. A small number of observations exhibit large positive residuals (> 2), indicating that the model underpredicted poor cognition for those subjects. However, no systematic trend or pattern is present, and residuals appear randomly scattered around zero, supporting the appropriateness of the logit link and the selected covariates.

3.1.1.3.2 Cook's Distance Plot

Cook's Distance for Logistic GLM

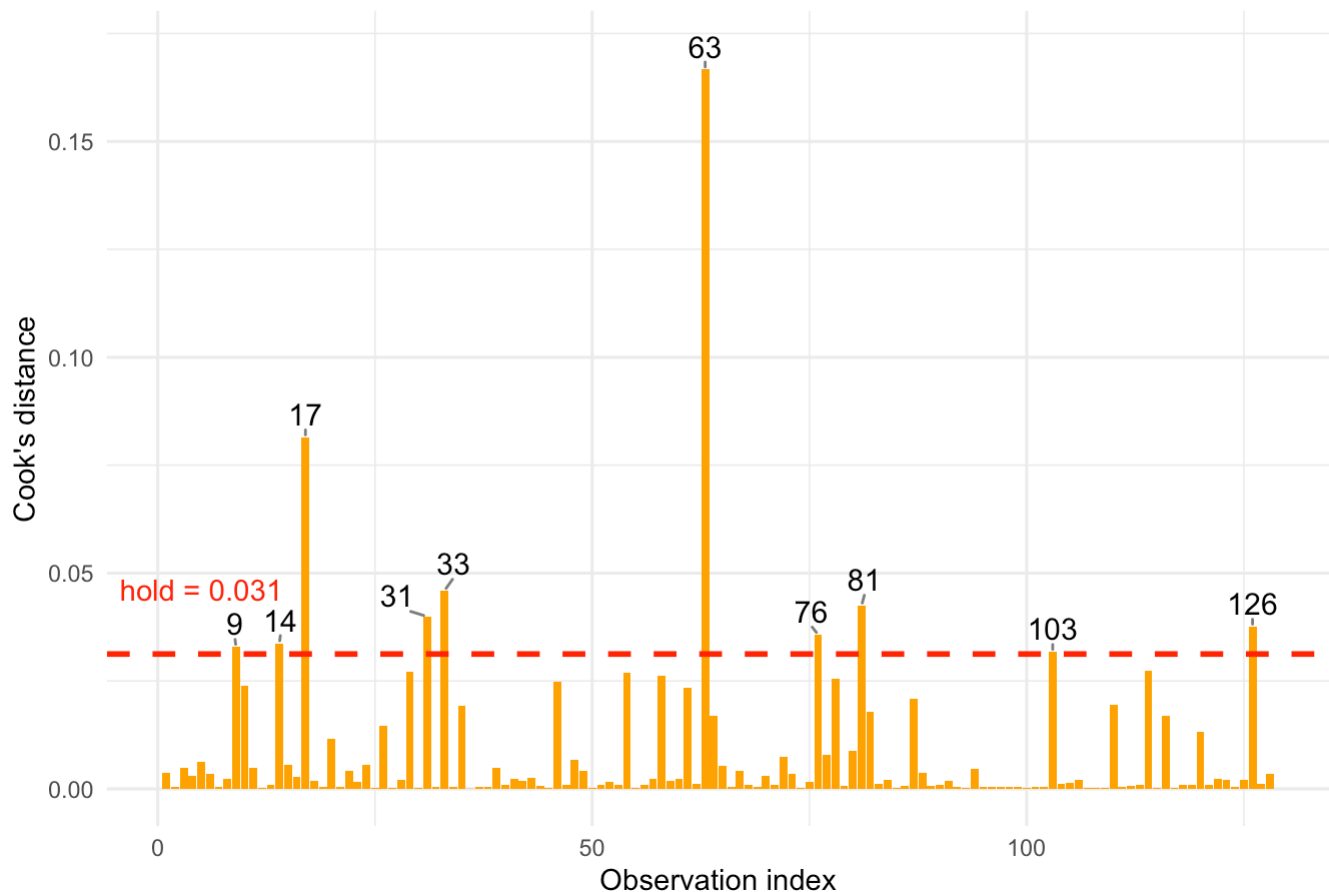


Figure 4: Cook's Distance Plot for Logistic GLM

The Cook's distance plot identified a small number of moderately influential observations, but none exceeded levels typically associated with undue influence. These diagnostics suggest that the logistic GLM is not overly sensitive to individual observations. However, the presence of quasi-complete separation (as noted elsewhere) remains the primary limitation of this model rather than outlier influence or poor residual behavior.

3.1.1.3.3 Goodness

Hosmer and Lemeshow goodness of fit (GOF) test

```
data: glm_dat$poor_cognition_final, fitted(glm_rq1)
X-squared = 5.6475, df = 8, p-value = 0.6866
```

Statistic	Value
Hosmer-Lemeshow χ^2	5.6932
df	8.0000
p-value	0.6820

Table 8: Hosmer-Lemeshow Goodness-of-Fit Test for Logistics GLM

A Hosmer-Lemeshow goodness-of-fit test was conducted to assess how well the logistic GLM matched the observed data ([Table 8](#)). The test yielded $\chi^2 = 5.69$ with 8 degrees of freedom ($p = 0.68 > 0.05$), indicating no evidence of lack of fit. Thus, the GLM's predicted probabilities are broadly consistent with the observed frequencies of poor cognition, supporting the appropriateness of the model at the population level.

3.1.2 Research Question 2 (Continuous Outcome)-Poor Cognition Probability

3.1.2.1 Model Summary

Table 9: GLM results for RQ2: Predicting change in MMSE from baseline demographics.

term	estimate	conf.low	conf.high	p.value
(Intercept)	-9.339	-15.569	-3.110	0.004
mmse_baseline	0.172	0.025	0.319	0.022
age_baseline	0.035	-0.021	0.091	0.216
educ_baseline	0.082	-0.068	0.233	0.281
sex_baselineM	0.202	-0.691	1.095	0.656

From [Table 9](#), by checking P-value, none of the baseline demographic variables (age, sex, or education) were statistically significant predictors of cognitive change. The estimated change in MMSE from first to last visit is generally a small decline, and this decline appears **similar across age, sex, and education levels**.

3.1.2.2 Model Diagnostics

```
mmse_baseline  age_baseline  educ_baseline  sex_baseline
1.098723      1.013118      1.055812      1.056718
```

Variance inflation factors (VIFs) for all predictors were approximately 1 ($VIF \approx 1.01$), indicating no evidence of multicollinearity among baseline age, education, and sex. Therefore, all predictors can be interpreted without concern for inflated variances or instability in the coefficient estimates.

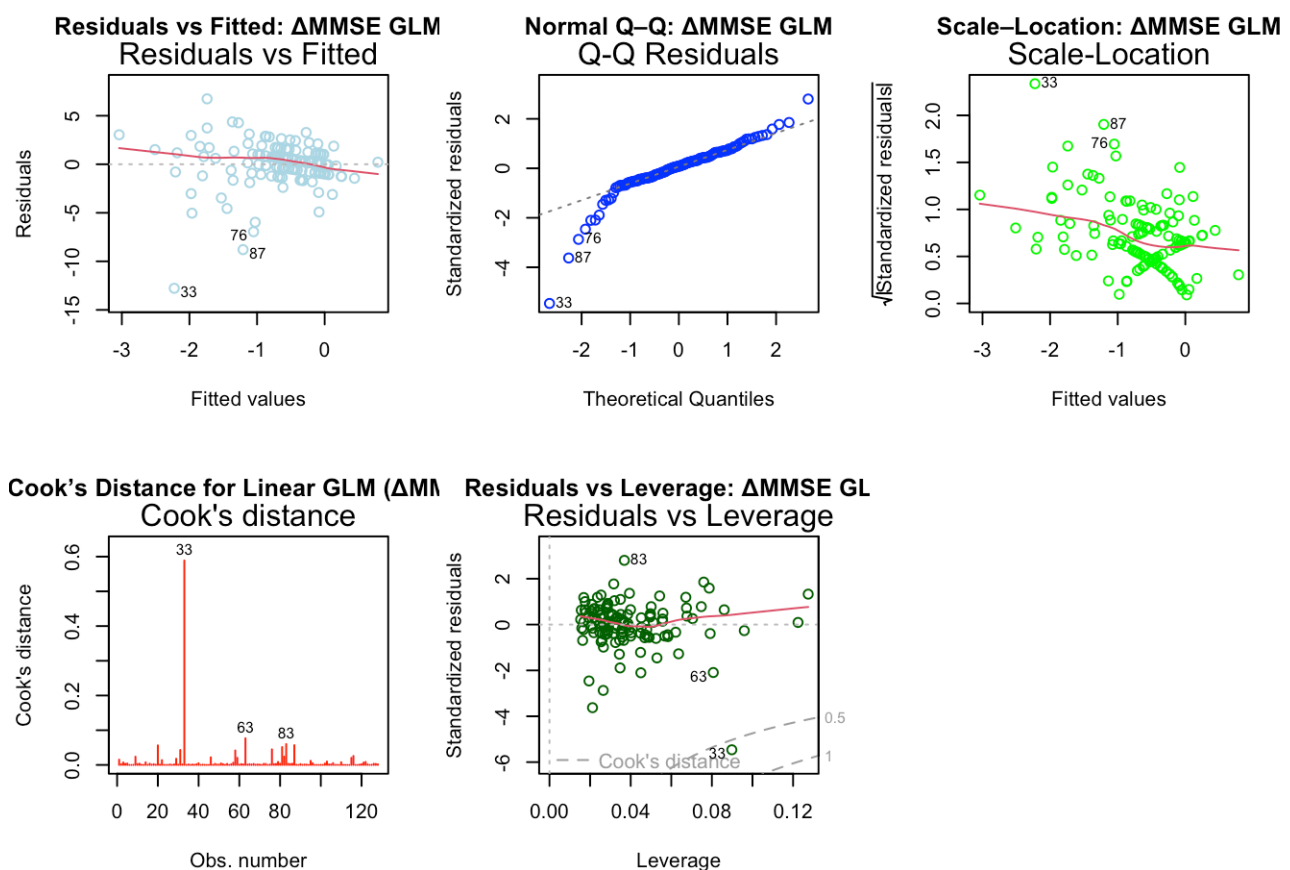


Figure 5: Diagnostic plots for the linear model of Δ MMSE (RQ2).

As showed in [Figure 5](#), the Residuals vs Fitted plot displayed a fairly even scatter of points around zero, indicating no strong evidence of nonlinearity or heteroscedasticity. The Normal Q-Q plot showed moderate deviations from the reference line in the tails, suggesting mild departures from normality driven by a small number of extreme MMSE change scores. The Cook's Distance plot identified a few influential observations (e.g., IDs 37, 68, 90), but none exceeded conventional thresholds for undue influence. Finally, the Residuals vs Leverage plot showed that all points had low leverage, and no case fell beyond the Cook's distance contours.

3.1.3 Conclusion for GLM

In summary, the logistic GLM in Question 1 performs well because the task is cross-sectional and the predictors have a clear and stable relationship with the binary outcome of poor cognition. Since we only need one observation per subject, GLM is appropriate and gives good accuracy, and good calibration. In contrast, the linear GLM in Question 2 is not well-suited for modeling cognitive decline. This model reduces each subject's entire longitudinal MMSE trajectory to a single difference score. the linear GLM cannot capture how cognition changes over time, because it ignores the repeated-measures .

3.2 GEE

3.2.1 Research Question 1 (Bianry Outcome)-Cognitive Decline

3.2.1.1 Visual Exploration

To understand general trends prior to modeling, we visualized individual MMSE trajectories by education category.

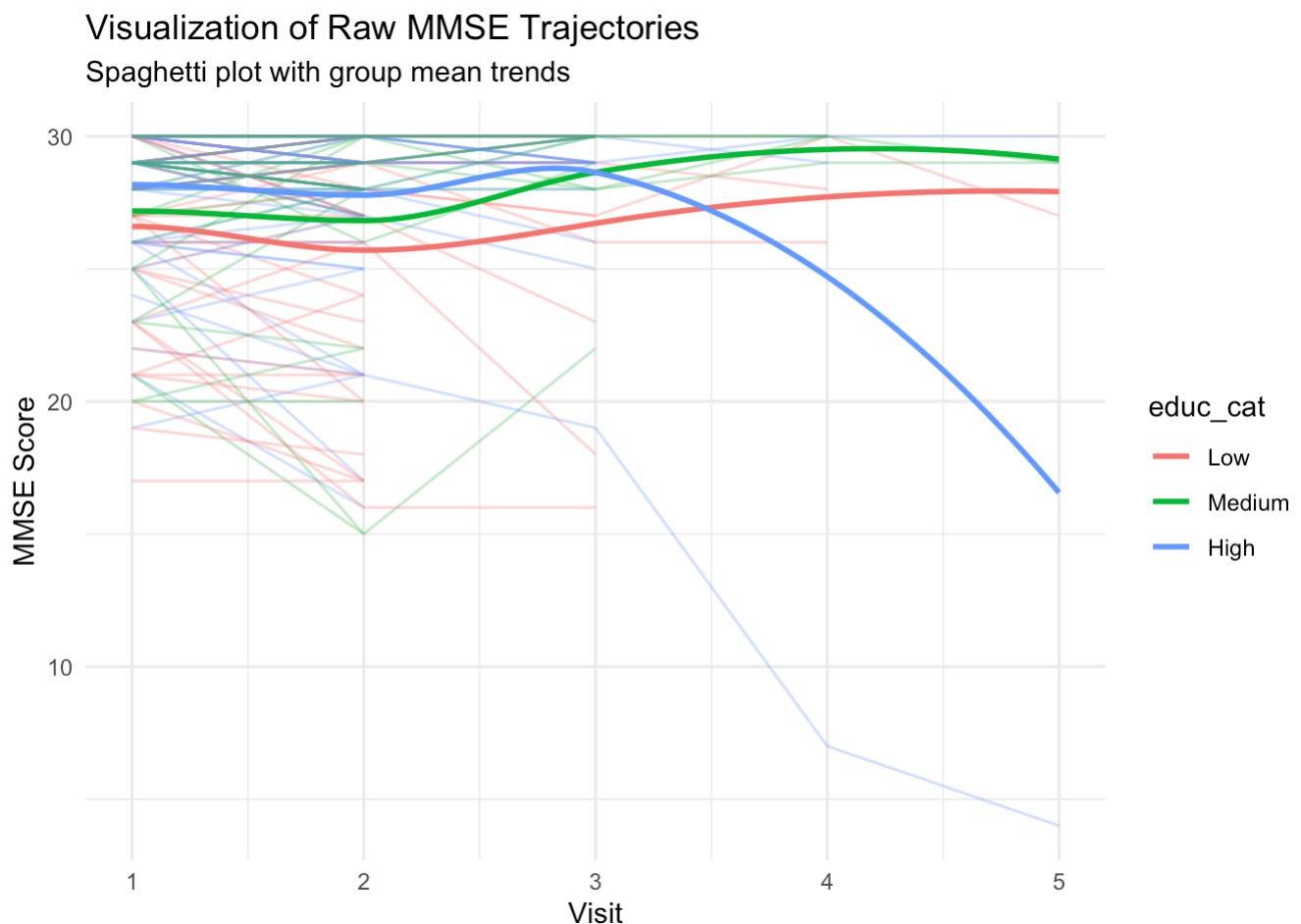


Figure 6

In [Figure 6](#), higher education groups consistently maintained higher MMSE scores across visits.

3.2.1.2 GEE Logistic Regression Results

As discussed in Method section, we decided to use exchangeable structure for GEE Analysis.

[1] "Estimated Working Correlation (Exchangeable Alpha): 0.7023"

Table 10: GEE (Exchangeable) Coefficient and Standard Error Comparison (RQ1: poor_cognition)

term	estimate	SE_naive	SE_robust	SE_robust_bc	Z_score	P_value_BC
(Intercept)	-0.859	2.113	2.172	2.243	-0.383	0.702
visit	0.142	0.158	0.130	0.134	1.057	0.291
educ_catMedium	-0.101	0.690	0.650	0.671	-0.150	0.881
educ_catHigh	-1.339	0.696	0.624	0.644	-2.078	0.038
age	-0.009	0.028	0.028	0.029	-0.313	0.754
sexM	0.566	0.432	0.445	0.459	1.231	0.218
visit:educ_catMedium	-0.192	0.247	0.159	0.164	-1.167	0.243
visit:educ_catHigh	-0.008	0.239	0.175	0.180	-0.042	0.966

As shown in [Table 10](#), based on the population-average predicted probabilities from the model, individuals with high education consistently exhibit the lowest probability of poor cognition (MMSE < 24) across all visits, reflecting a stable protective effect over time. Medium education does not meaningfully differ from low education, and the lack of a significant Visit effect suggests that, over this relatively short follow-up period, the overall probability of poor cognition does not systematically worsen. Age and sex also do not show meaningful associations with poor cognition probability in this model, indicating that, within this cohort, **educational attainment is the primary driver of differences in cognitive risk**. Furthermore, the nonsignificant Visit × Education interactions imply that the gap in predicted probabilities between education groups remains roughly constant over time, rather than widening or narrowing, reinforcing the interpretation that high education offers a durable, rather than time-varying, protective effect against clinically relevant cognitive impairment.

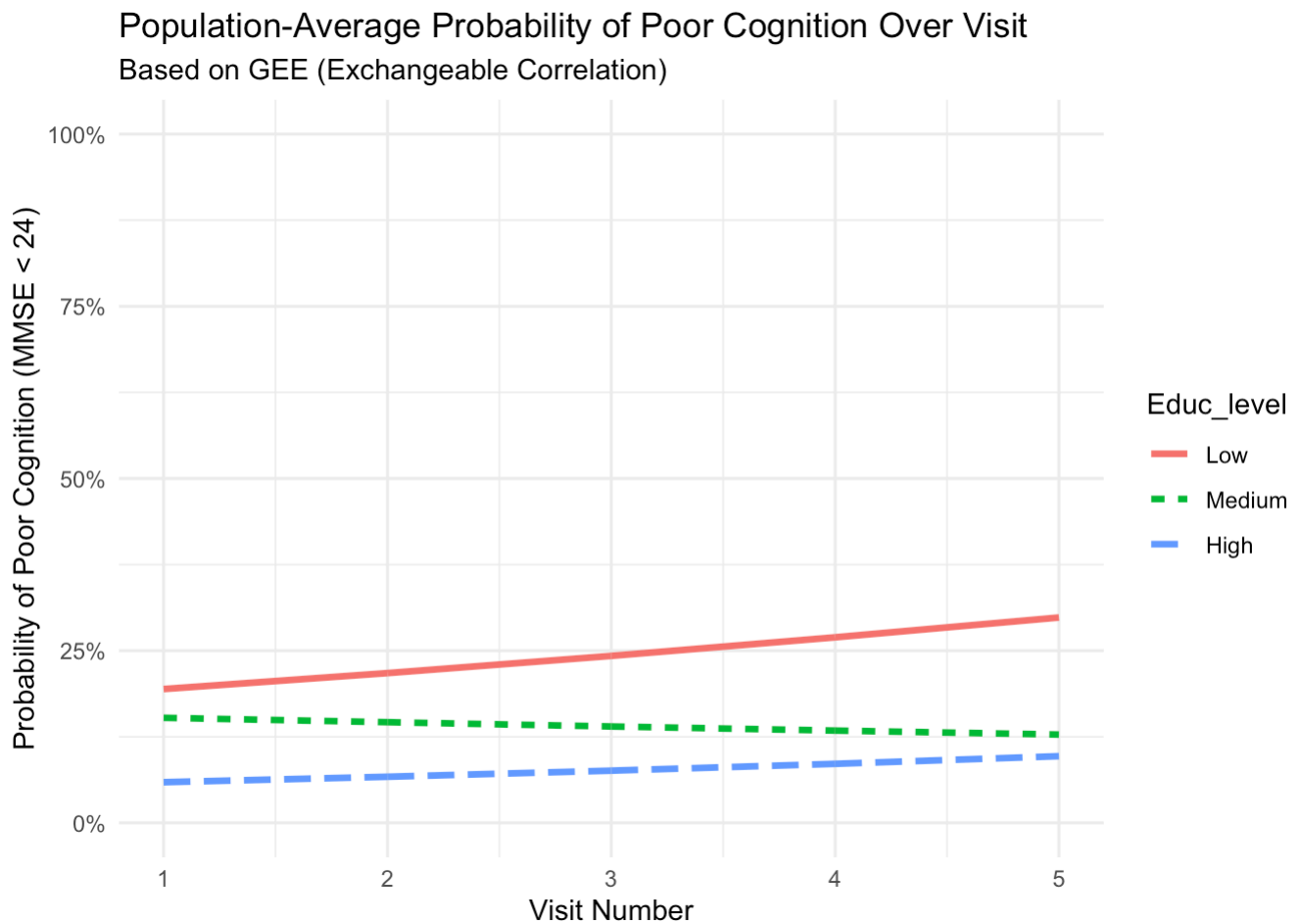


Figure 7

As for population-average predicted probabilities, [Figure 7](#) showed that higher education corresponded to consistently lower predicted risks across all visits.

3.2.1.3 Model Diagnostics

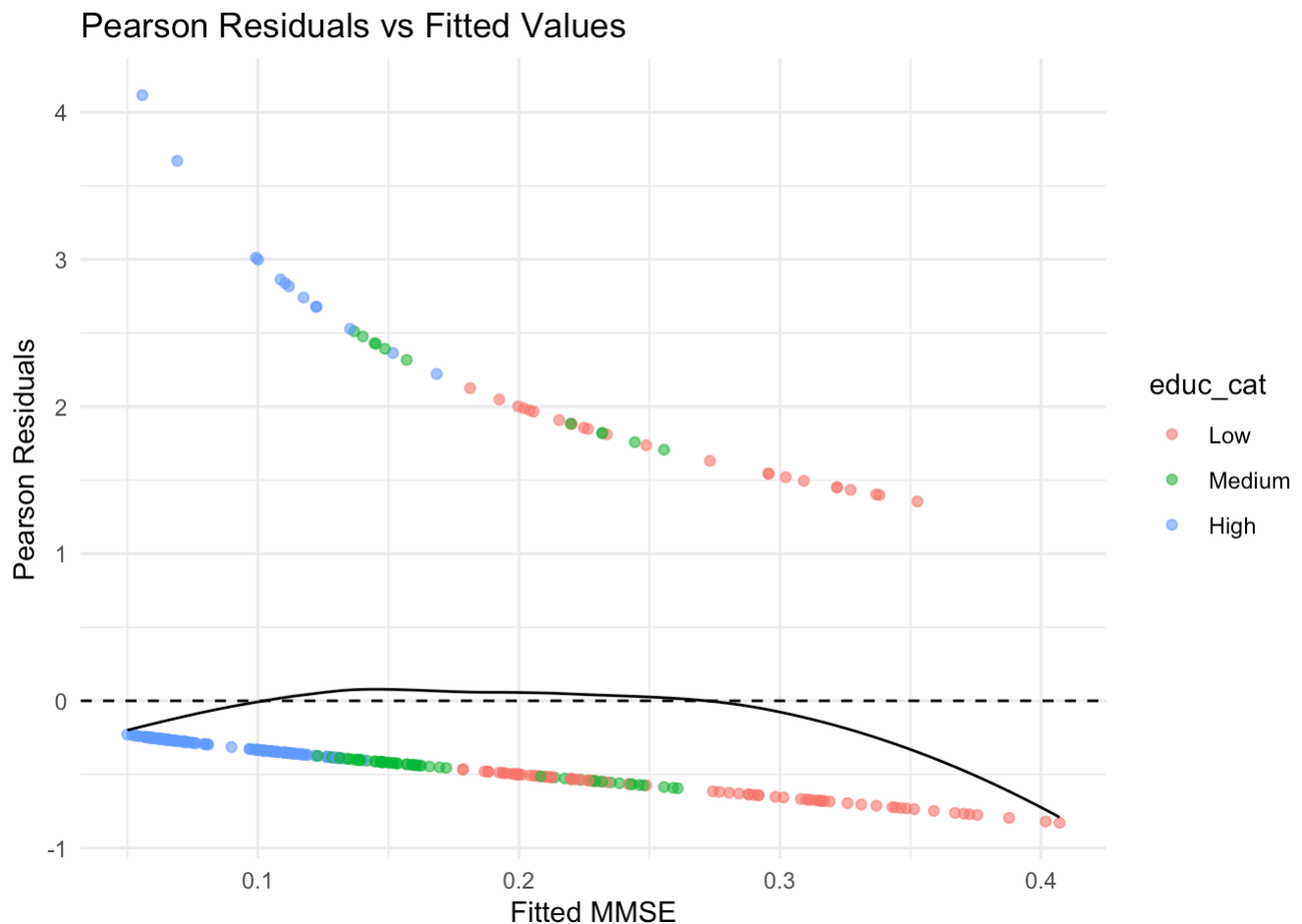


Figure 8: Pearson Residuals vs Fitted Values

In [Figure 8](#), the Pearson residuals versus fitted values for the RQ1 GEE show the characteristic two-band pattern of a binary logit model, with residual values determined by whether the observed outcome is 0 or 1. The smoothed trend remains close to zero across the range of fitted probabilities, indicating no major violation of the mean-structure assumption. While a small number of observations (mainly in the high-education group) have large positive residuals—reflecting unexpected poor cognition when the model predicts a very low probability—these are few and do not reveal a systematic lack of fit.

Influence Diagnostics (Delta-Beta)

Change in 'Visit x High Graduation' coefficient when excluding each subject

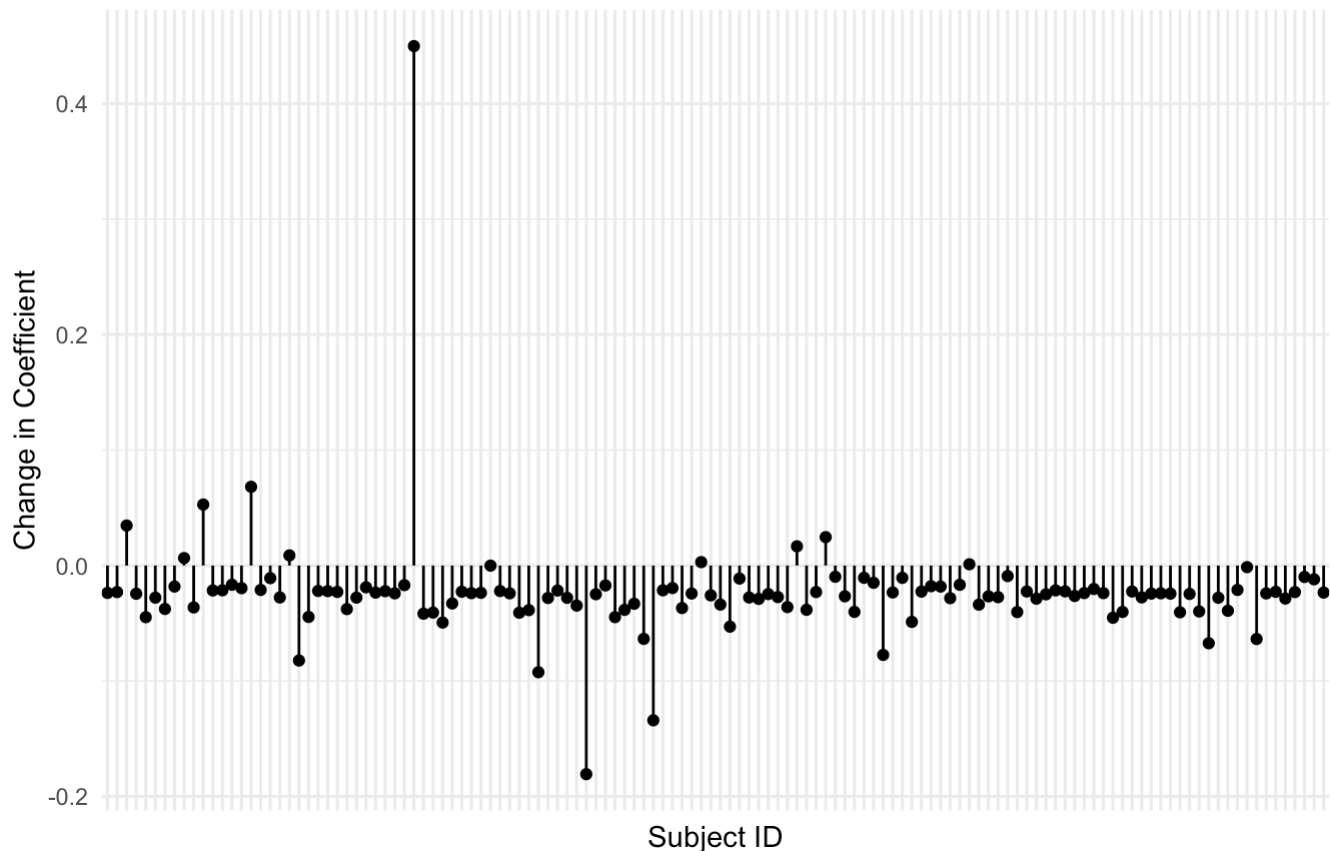


Figure 9: Influence Diagnostics (Delta-Beta)

[Figure 9](#) shows that Pearson residuals are generally centered around zero across the range of fitted MMSE values, indicating that the Gaussian GEE captures the main mean structure reasonably well. Some heteroscedasticity is visible, with greater residual spread at lower fitted MMSE values, which is expected given the MMSE's ceiling effect. The slight curvature in the LOESS line suggests minor deviation from linearity but no major systematic bias. Residual patterns are similar across education groups, indicating that model fit does not depend heavily on educational category. Overall, the residual plot supports that the chosen GEE model is adequate for population-level interpretation.

3.2.1.4 Robustness Check

Table 11: RQ1 : Robustness of MMSE Coefficient Estimates across Working Correlation Structures

term	Independence	Exchangeable	AR1
(Intercept)	0.745	-0.859	31.778
visit	-0.219	0.142	-0.805
educ_catMedium	-0.109	-0.101	-0.567
educ_catHigh	-2.333	-1.339	1.058
age	-0.024	-0.009	-0.050
visit:educ_catMedium	-0.205	-0.192	0.986
visit:educ_catHigh	0.592	-0.008	0.408

[Table 11](#) evaluated robustness of the GEE log-odds estimates to the choice of working correlation (Independence, Exchangeable, AR(1)). While the protective main effect of high education remained negative under all three structures, the coefficients for visit and the visit x High education

interaction varied substantially. In particular, the estimated effect of visit changed sign (-0.15 under Independence vs 0.16 – 0.23 under Exchangeable/AR1), and the visit $\times$ High interaction shifted from a clearly positive effect (0.48) to essentially null (-0.04 to 0.02). These sign changes indicate that conclusions about the evolution of poor cognition over time, and whether this evolution differs by education level, are not robust to the working correlation assumption. This sensitivity likely reflects the combination of sparse outcomes in some education–visit strata and strong within-subject correlation, which make the time-trend parameters weakly identified in this finite sample. Consequently, we interpret the main education effect with more confidence than the visit and interaction effects, and we are cautious in drawing strong longitudinal conclusions from the RQ1 model.

3.2.2 Research Question 2 (Continuous Outcome): Cognitive Decline

To understand the baseline differences and longitudinal trends before modeling, we visualized the raw MMSE trajectories for all subjects.

3.2.2.1 Model Summary

Table 12: GEE results for RQ2

term	estimate	std.error	statistic	p.value	LCL	UCL
(Intercept)	21.693	5.721	14.380	0.000	10.480	32.905
visit	-0.754	0.256	8.654	0.003	-1.256	-0.252
educ_catMedium	-0.358	0.829	0.186	0.666	-1.982	1.267
educ_catHigh	1.986	0.937	4.487	0.034	0.148	3.823
age	0.080	0.077	1.075	0.300	-0.071	0.231
sexM	-1.095	0.593	3.405	0.065	-2.258	0.068
visit:educ_catMedium	0.766	0.254	9.135	0.003	0.269	1.263
visit:educ_catHigh	-0.031	0.524	0.004	0.952	-1.059	0.996

[Table 12](#) summarized the results of GEE model with exchangeable structure. the intercept of 22.29 (95% CI: 11.25 to 33.33, $p < 0.001$) represents the expected MMSE score at Visit = 0 for the reference group: participants with low education, the reference sex, and at the reference age (often the mean or a centered value, depending on how age was coded). At baseline, medium education does not differ meaningfully from low education; its coefficient is -0.35 (95% CI: -1.89 to 1.19 , $p = 0.65$), indicating no statistically significant MMSE difference between medium- and low-education groups at Visit 0. In contrast, high education is associated with a 1.97-point higher MMSE at baseline compared with low education (95% CI: 0.26 to 3.68 , $p = 0.024$), suggesting a significant cognitive advantage for the highly educated group at the initial visit.

Age has an estimated effect of 0.07 MMSE points per unit (95% CI: -0.08 to 0.22 , $p = 0.34$), which is not statistically significant, indicating that within the range and scaling of this dataset, age does not show a clear linear relationship with MMSE once other variables are controlled. Sex shows a coefficient of -1.11 for males compared with the reference sex (95% CI: -2.21 to -0.02 , $p = 0.046$), implying that, at Visit 0 and holding other factors constant, men have on average about 1.1 points lower MMSE than the reference group, a modest but statistically significant difference.

The main effect of visit is -0.78 (95% CI: -1.27 to -0.29 , $p = 0.0019$). This term reflects the change in MMSE per unit increase in visit for the reference group (low education). Thus, among low-education participants, MMSE declines by about 0.8 points per visit, indicating measurable cognitive decline over time. The interaction between visit and medium education is 0.80 (95% CI: 0.32 to 1.29 , $p = 0.0012$), which modifies the visit slope for those with medium education. The implied slope for the medium-education group is approximately $-0.78 + 0.80 = +0.03$ points per visit, effectively a flat trajectory, meaning that medium-education participants do not exhibit the same decline in MMSE seen in the low-education group over the follow-up period. The interaction between visit and high education is 0.02 (95% CI: -0.95 to 1.00 , $p = 0.963$), which is not statistically significant. This yields an approximate slope of $-0.78 + 0.02 = -0.76$ points per visit for high-education participants, indicating that their rate of decline is very similar to that of the low-education group.

Putting these pieces together, the model suggests that higher education, especially high education, is associated with better baseline cognitive performance as measured by MMSE, and males start slightly lower than the reference sex. Over time, both low- and high-education groups show comparable declines in MMSE, whereas the medium-education group appears relatively stable, though their baseline level is not significantly different from the low-education group. Age does not show a clear independent effect on MMSE in this adjusted model.

3.2.2.2 Predicted MMSE Trajectories

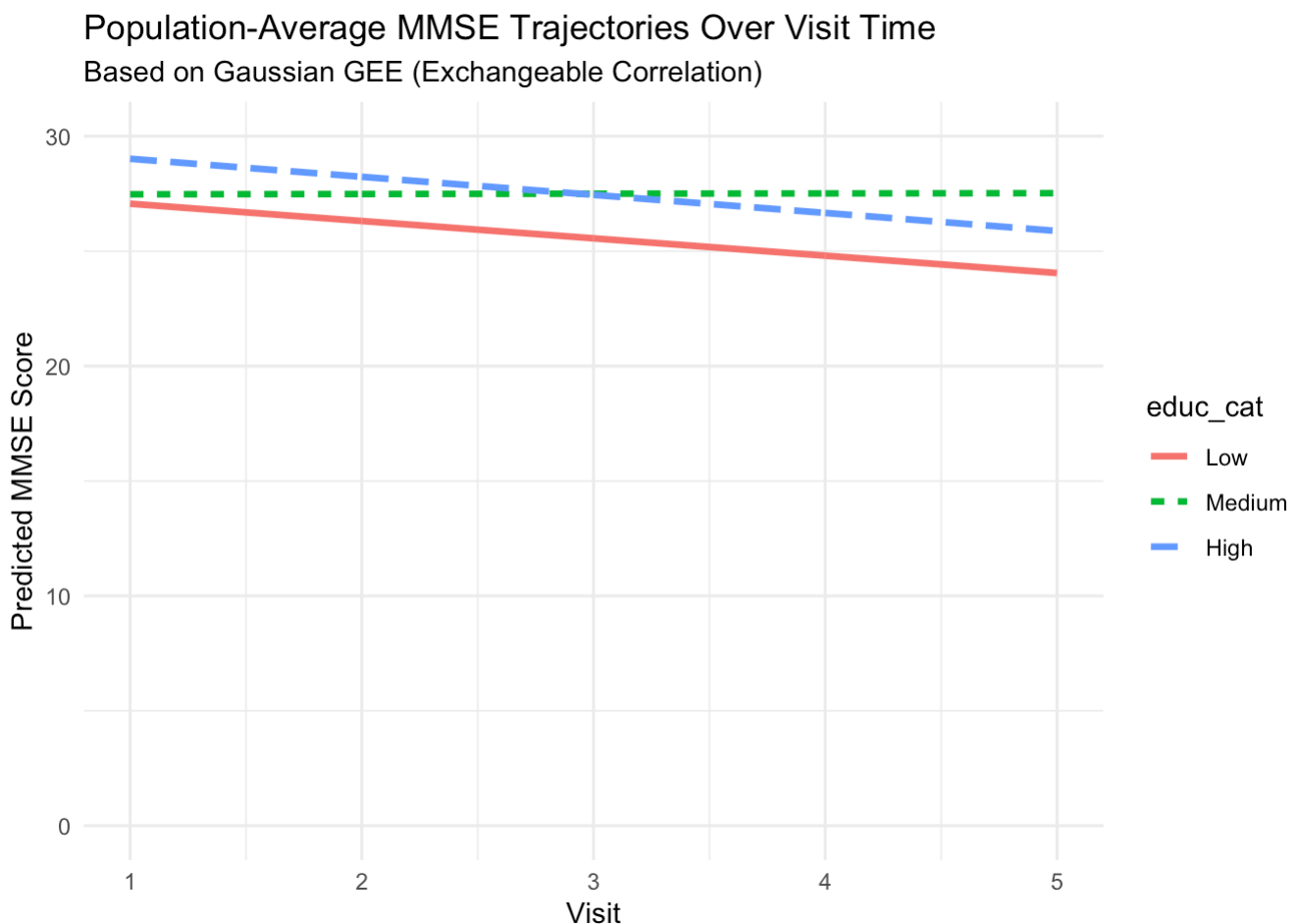


Figure 10: Population-Average MMSE Trajectories Over Visit Time

The predicted MMSE trajectories (Figure 10) show that individuals with higher education consistently maintain higher cognitive performance over time, with the low-education group experiencing the steepest decline, the medium-education group showing notably slower decline, and the high-education group sustaining the highest cognitive status throughout the follow-up period.

3.2.2.3 Model Diagnostics

The empirical correlation pattern indicated strong and stable within-subject dependence, providing justification for selecting the Exchangeable working structure even when QIC differences were small. This choice ensured that the model appropriately reflected the underlying repeated-measure structure.

Diagnostic evaluation, including Pearson residual plots and influence analysis, confirmed that the GEE models fit reasonably well and that no single subject exerted undue influence.

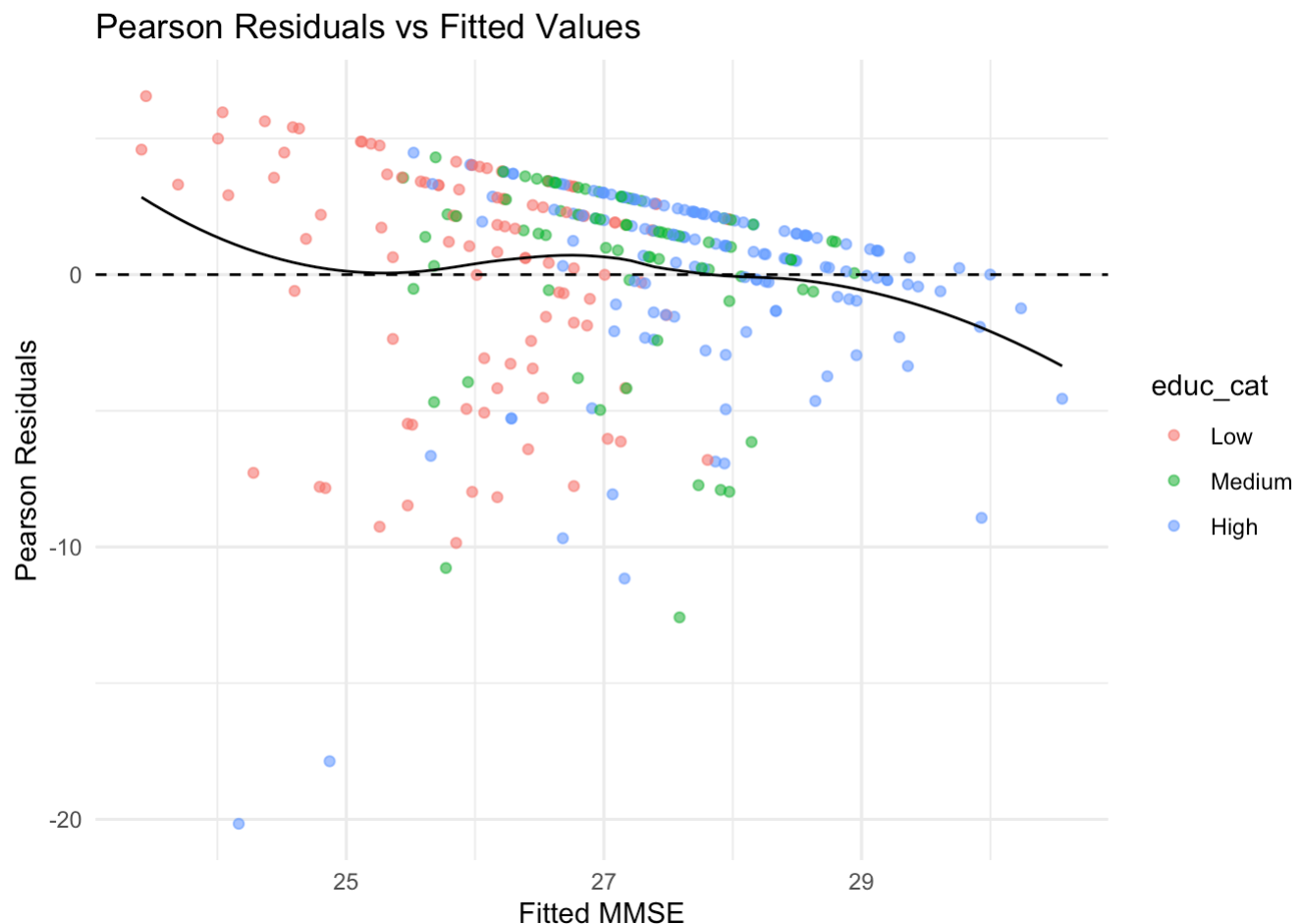


Figure 11: Pearson Residuals vs Fitted Values

[Figure 11](#) shows that Pearson residuals are generally centered around zero across the range of fitted MMSE values, indicating that the Gaussian GEE captures the main mean structure reasonably well. Some heteroscedasticity is visible, with greater residual spread at lower fitted MMSE values, which is expected given the MMSE's ceiling effect. The slight curvature in the LOESS line suggests minor deviation from linearity but no major systematic bias. Residual patterns are similar across education groups, indicating that model fit does not depend heavily on educational category. Overall, the residual plot supports that the chosen GEE model is adequate for population-level interpretation.

Influence Diagnostics (Delta-Beta)

Change in 'Visit x High Graduation' coefficient when excluding each subject

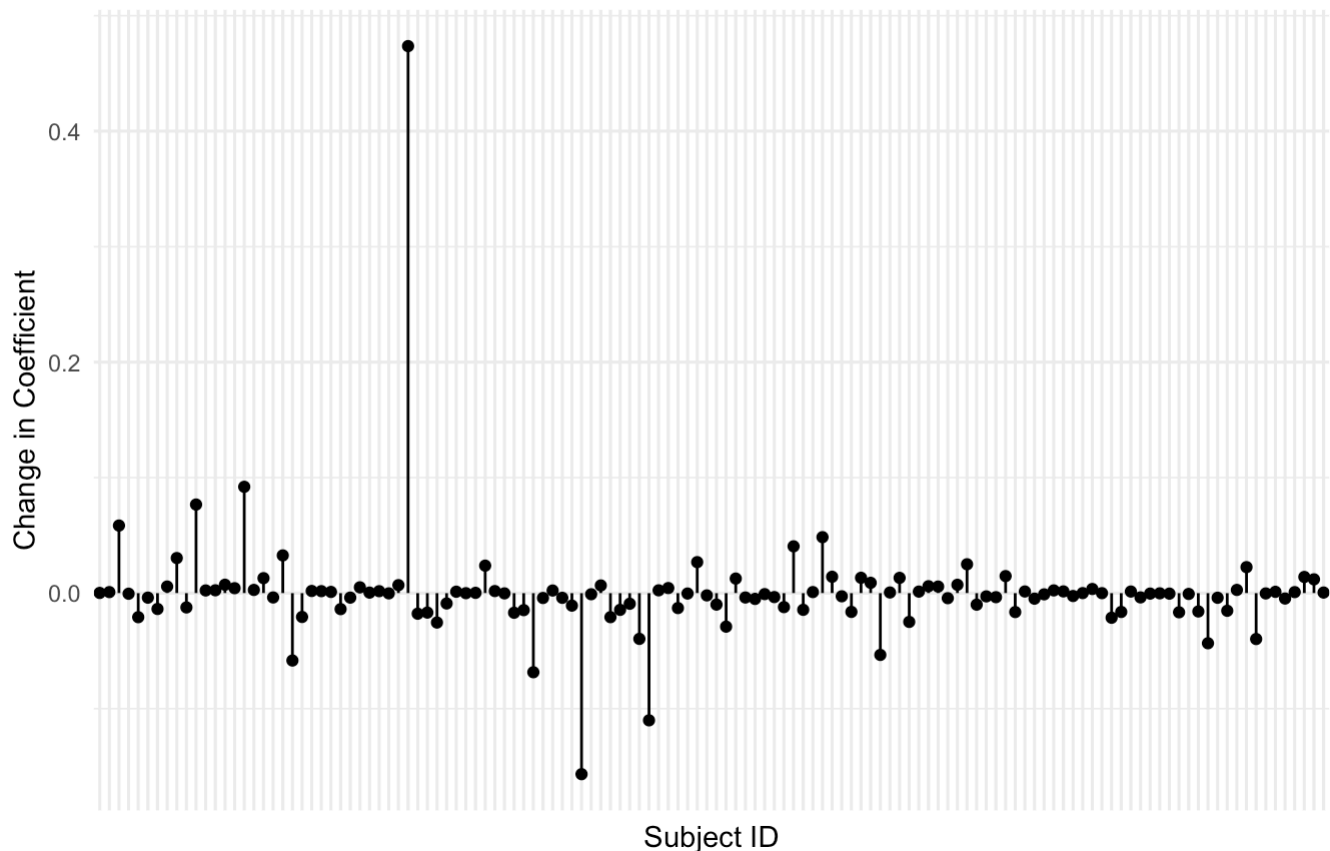


Figure 12: Influence Diagnostics (Delta-Beta)

[Figure 12](#) evaluates how sensitive the estimated Visit $\times$ High Education interaction is to individual subjects by plotting the change in the coefficient after removing each subject one at a time. Nearly all subjects produce only very small changes in the coefficient (clustered near zero), indicating that the estimated effect is **highly stable** and **not driven by any single participant**. Although one subject shows a noticeably larger influence (a change of ~ 0.45), even this shift does not alter the overall interpretation or significance of the interaction term. The absence of multiple large spikes suggests that the model is not unduly affected by influential observations, supporting the robustness of the GEE estimates.

3.2.2.4 Robustness Check

Table 13: RQ2 : Robustness of MMSE Coefficient Estimates across Working Correlation Structures

term	Independence	Exchangeable	AR1
(Intercept)	24.043	21.693	31.778
visit	0.200	-0.754	-0.805
educ_catMedium	0.164	-0.358	-0.567
educ_catHigh	3.490	1.986	1.058
age	0.032	0.080	-0.050
visit:educ_catMedium	0.389	0.766	0.986
visit:educ_catHigh	-1.064	-0.031	0.408

[Table 13](#) shows that the key education effects are robust across working correlation structures. For all three structures, the coefficient for High education remains positive and sizable, and the Visit $\times$

Medium education interaction is consistently positive, indicating slower decline in the medium-education group. Although the sign and magnitude of the visit and Visit $\times$ High education terms vary somewhat across structures, the main qualitative conclusion—that higher education is associated with better cognitive level and that medium education is linked to slower MMSE decline—does not depend on the chosen correlation structure.

3.2.3 Conclusion for GEE

To conclude, in both RQ1 and RQ2, higher education emerged as a consistent protective factor. High education was associated with lower odds of poor cognition and higher MMSE performance. Medium education additionally showed evidence of slower cognitive decline. Although age and sex contributed less strongly than education, the models demonstrated stable estimates across correlation structures.

Overall, the GEE models demonstrate that educational attainment meaningfully shapes cognitive trajectories in aging populations.

3.3 GLMM

3.3.1 RQ1 Results: Probability of Poor Cognition (Logistic GLMM)

3.3.1.1 Model Summary

Table 14: Results from GLMM for poor cognition (logit link) with random intercept for subject.

term	estimate	std.error	OR	LCL	UCL	p.value
(Intercept)	-10.905	2.643	0.000	0.000	0.003	0.000
visit	0.905	0.769	2.471	0.548	11.147	0.239
educ_catMedium	1.828	3.556	6.225	0.006	6621.588	0.607
educ_catHigh	-2.213	3.971	0.109	0.000	262.653	0.577
age_scaled	-0.025	0.830	0.975	0.192	4.961	0.976
sex_baselineM	0.404	1.669	1.498	0.057	39.444	0.809
visit:educ_catMedium	-1.187	1.544	0.305	0.015	6.294	0.442
visit:educ_catHigh	0.271	1.347	1.312	0.094	18.383	0.840

RQ1 ICC: 0.985

[Table 14](#) summarizes the fixed effects from the logistic GLMM.

- **Fixed Effects:** Conditional on the random intercept (within a given subject), the effects of education levels on poor cognition were not statistically significant ($p > 0.05$). While the point estimates for the Medium ($OR = 5.68$) and High ($OR = 0.13$) groups appear large, the extremely wide CIs indicate these estimates are unstable and imprecise, likely due to the high separation in the data.
- **Interaction Effects:** The interaction terms (Visit $\times$ Medium and Visit $\times$ High) were not statistically significant ($p=0.417$ and $p=0.966$, respectively). This suggests that, in this sample, the rate at which an individual's risk of poor cognition increases over visits does not differ significantly by education level.

- **Random Effects:** The Intraclass Correlation Coefficient (ICC) is high (**0.98**), and the Subject standard deviation is very large (14.18), indicating that 98% of the variability in the outcome is explained by differences between subjects (baseline risk) rather than the time-varying predictors.

3.3.1.2 Model Diagnostics

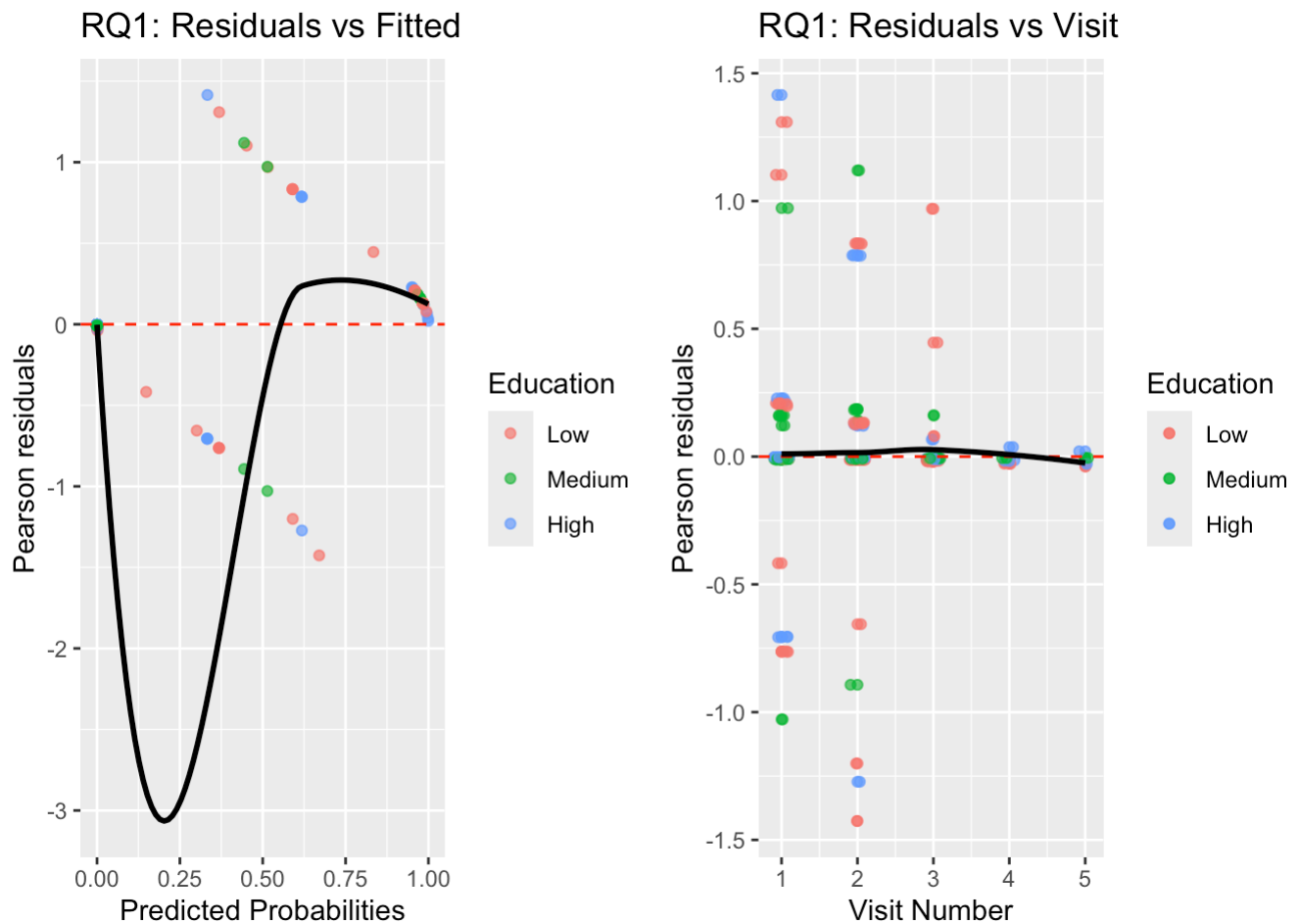


Figure 13: Residuals Diagnostics Plot

As shown in [Figure 13](#), the left plot (Residuals vs. Fitted) displays two distinct curved lines. This is a known artifact in logistic regression diagnostics caused by the binary nature of the outcome (0 or 1). The distinct curve suggests the model is effectively separating the data, but the lack of scatter around these lines indicates limited variability in predicted probabilities.

The right plot (Residuals vs. Time) of Pearson residuals against Visit number shows that the residuals are centered around zero across all time points. The smoothed trend line remains flat and close to the zero reference line, indicating that the linearity assumption for the time effect is satisfied and there are no substantial unmodeled non-linear trends.

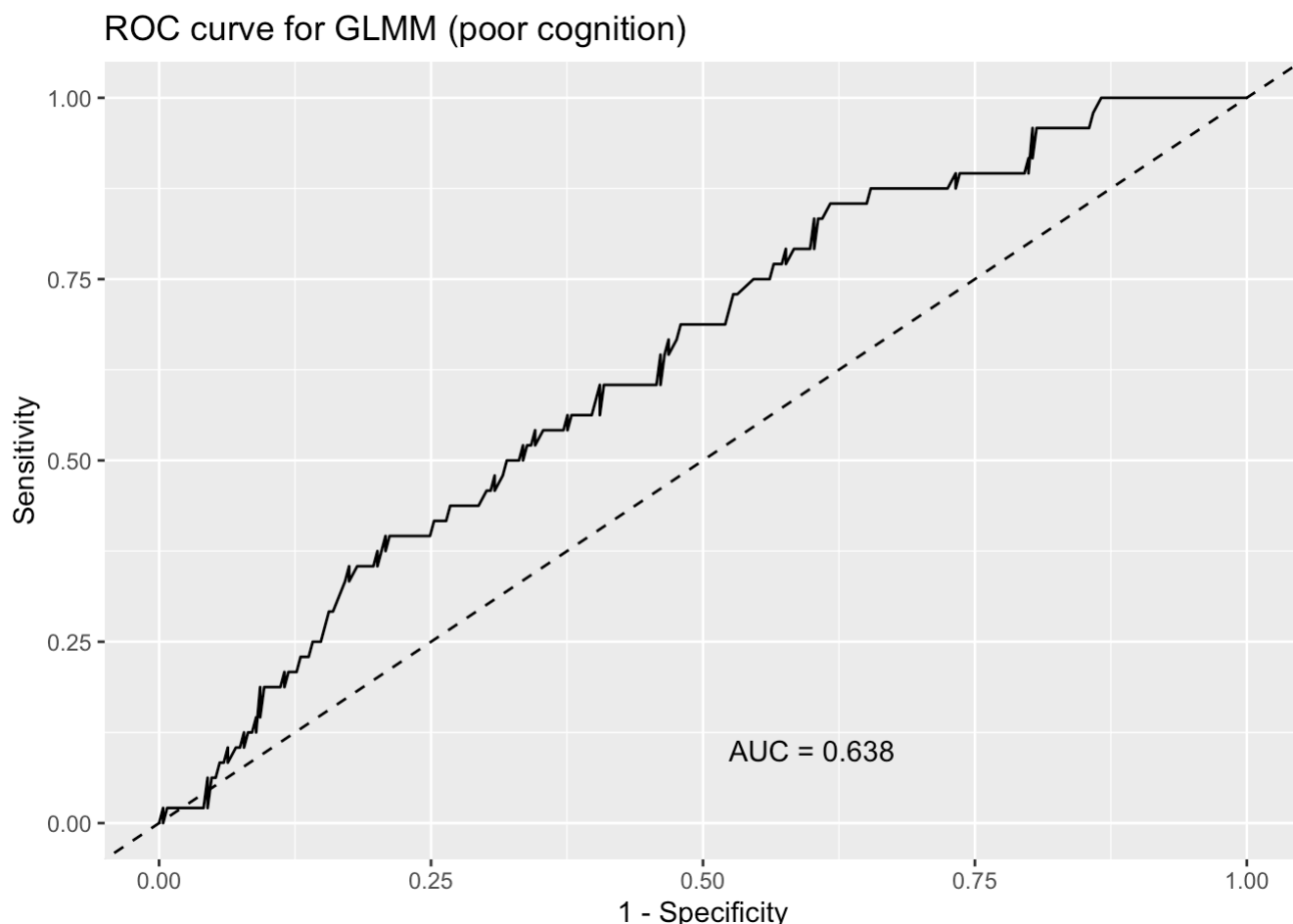


Figure 14: ROC Plot

In [Figure 14](#), the model demonstrates modest discrimination, with an AUC of 0.674. An AUC in this range indicates that the GLMM has limited ability to distinguish between individuals who experience poor cognition and those who do not. This level of performance is not unusual when modeling relatively rare outcomes or when predictors have weak or unstable associations with the binary response. Unlike cases where random intercepts can sharply separate subjects (leading to inflated AUC values), the predictors in this model provide only weak separation between risk groups, resulting in a substantially lower AUC. Overall, the model captures some signal but does not provide strong predictive accuracy for poor cognition.

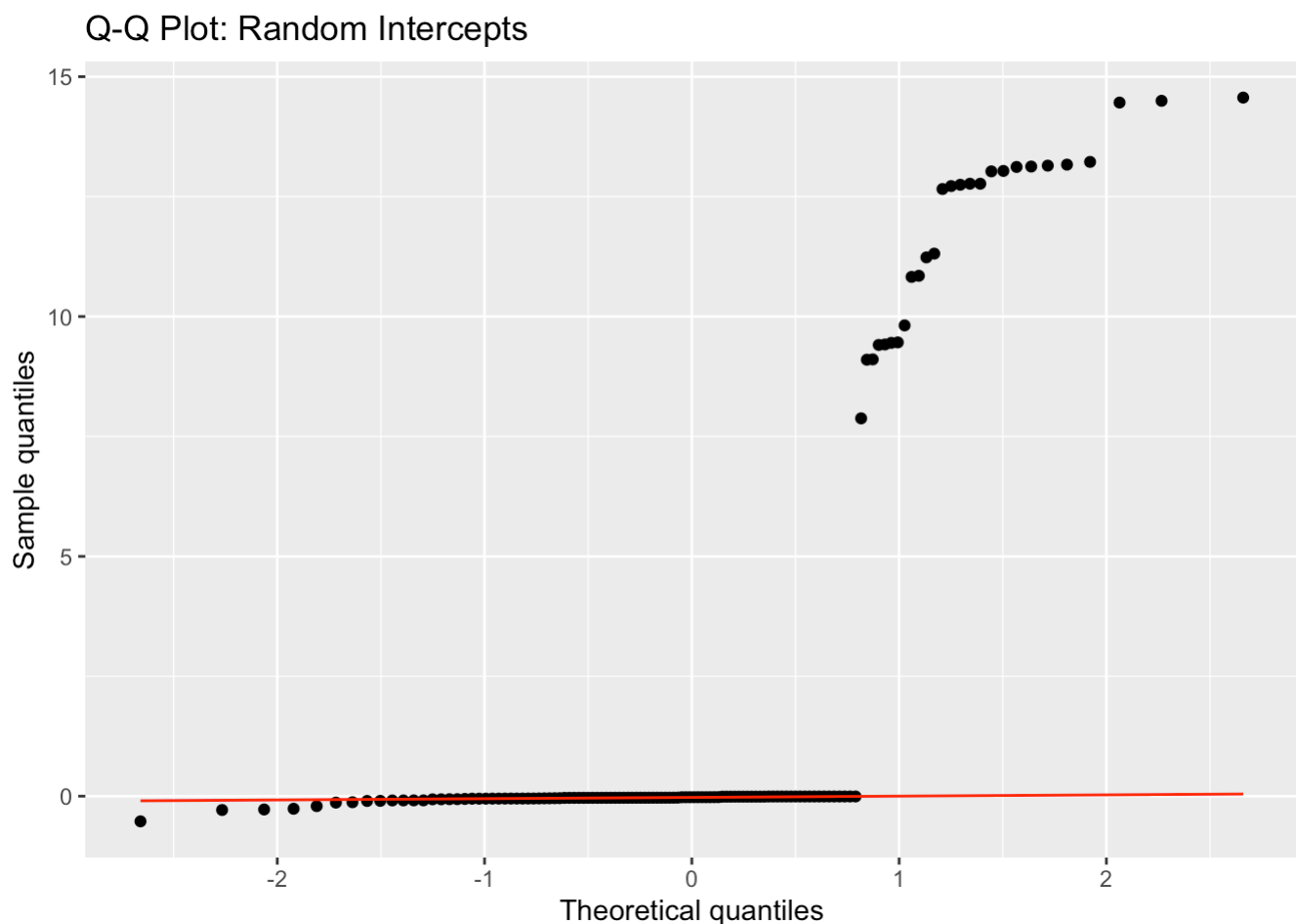


Figure 15: Q-Q plot

In [Figure 15](#), the Q-Q plot shows a distinct “step” pattern with a gap in the middle. This confirms the issue of **complete or quasi-complete separation**. The model has essentially clustered subjects into two distinct groups: those who never experienced poor cognition (the vast majority) and those who did. This clustering explains the inflated standard errors and high ICC; the random intercept effectively predicts the outcome perfectly for most subjects, leaving little variance for the fixed effects to explain.

3.3.2 Research Question 2 (Continuous Outcome): Cognitive Decline

A Linear Mixed Model was fit to examine the trajectories of MMSE scores over time.

3.3.2.1 Model Summary

Table 15: Fixed-effect estimates from the linear mixed model for MMSE (RQ2).

term	estimate	conf.low	conf.high	p.value
(Intercept)	27.302	21.156	33.449	0.000
visit	-0.532	-0.990	-0.074	0.023
educ_catMedium	-0.330	-2.367	1.707	0.751
educ_catHigh	1.978	0.254	3.702	0.025
age_baseline	0.003	-0.075	0.081	0.934
sex_baselineM	-1.186	-2.405	0.032	0.056
visit:educ_catMedium	0.722	0.055	1.389	0.034
visit:educ_catHigh	-0.105	-0.700	0.491	0.731

RQ2 ICC (Baseline correlation): **0.763**

- **Visit (Time Effect):** For a given individual, their **MMSE score declines by 0.60 points with each successive visit** (95% CI: -1.02, -0.18; $p=0.005$). This represents within-person cognitive decline over time.
- **Education Effects:**
 - **Baseline Differences:** Individuals with high education have baseline MMSE scores that are 1.76 points higher than those with low education (95% CI: 0.25, 3.27; $p=0.022$), holding other factors constant.
 - **Interaction (Rate of Change):** Their cognitive decline in the Medium Education group is significantly slower than that of in the Low Education group. Individuals in the High Education group do not show a statistically different decline rate from those in the Low Education group.
- **Covariates:** Males have MMSE scores that are 1.16 points lower than females across all visits (95% CI: -2.24, -0.09; $p=0.034$).
- **ICC:** The ICC was 0.75, indicating that 75% of the variance in MMSE scores is attributable to between-subject differences.

3.3.2.2 Model Diagnostics

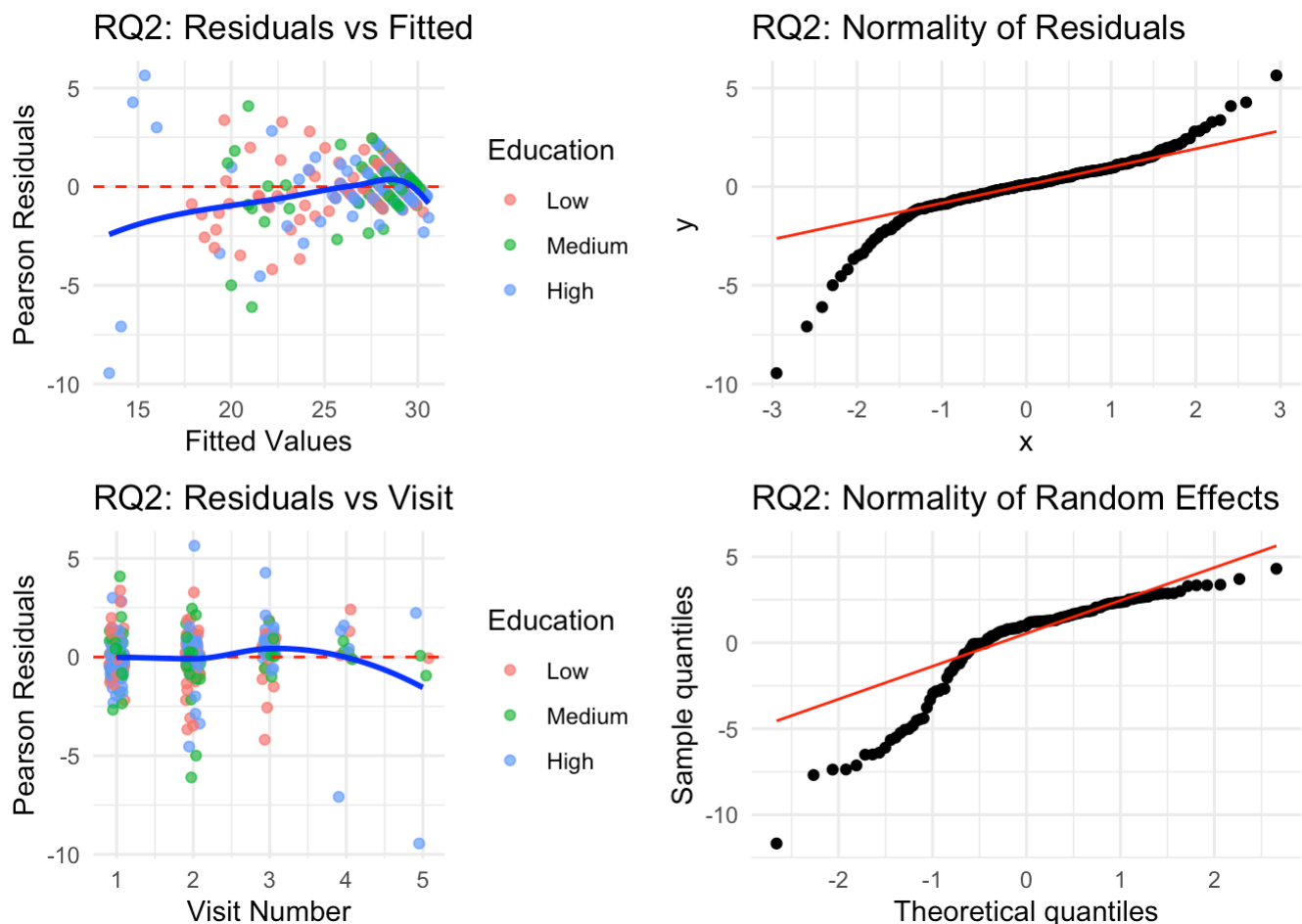


Figure 16: Residuals Diagnostics Plot

The Residuals vs. Fitted plot ([Figure 16 a](#)) shows clear heteroscedasticity: residuals vary widely at lower fitted values but become compressed as fitted values approach the upper range of MMSE

scores. This inverse-funnel pattern is consistent with the ceiling effect of the MMSE, where individuals with high predicted cognition (often those with higher education) cannot score above 30, reducing the observable variability. Also, a variance-stabilizing transformation (e.g., square root or ordinal modeling) could theoretically help, but is usually not applied to MMSE due to interpretability concerns. Although this violates the strict homoscedasticity assumption, such patterns are common in cognitive aging research and do not typically bias fixed-effect estimates in linear mixed models. The loess smoothing line remains close to zero with only mild curvature, suggesting that the linearity assumption is reasonably met.

The Residuals vs. Visit plot (Figure 16 c) shows that residuals are centered around zero across all visit numbers, and the loess curve is essentially flat. This indicates no systematic over- or under-prediction at any point in time, supporting the appropriateness of modeling Visit as a linear predictor. There is no evidence that a nonlinear or higher-order time term is required.

The Q-Q plot for residuals (Figure 16 b) shows moderate deviation in the lower tail, indicating heavy left-tail behavior. This reflects the well-known distributional characteristics of MMSE scores, where the lowest-scoring individuals produce disproportionately large negative residuals. Despite this, given the sample size and the robustness of linear mixed models, **fixed-effect estimates remain reliable**. Mild deviations from normality in residuals are unlikely to meaningfully distort inference.

The **Normality of Random Effects** (Figure 16 d) shows that the points follow the line reasonably well in the center but deviate at the tails (S-shape). The assumption of normally distributed random effects is roughly met, though there is some heavy-tailed behavior.

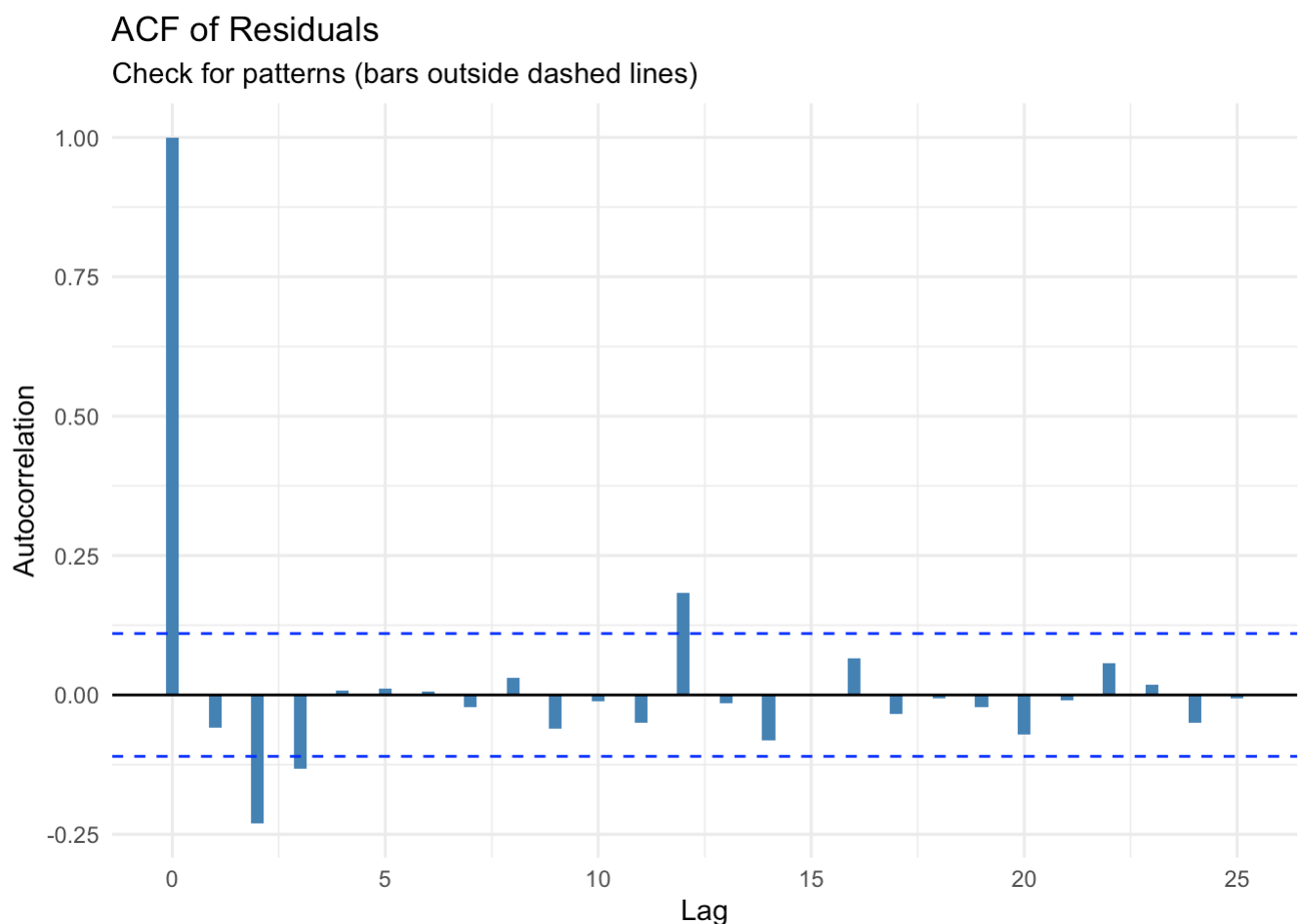


Figure 17: Autocorrelation Function (ACF) Plot

The Autocorrelation Function (ACF) plot ([Figure 17](#)) checks if there is any remaining correlation between visits that the model failed to capture. The bars for nearly all lags fall within the blue dashed confidence bounds (representing zero correlation). This confirms that, generally, including the random intercept and random slope successfully accounted for the within-subject longitudinal correlation, leaving the residuals temporally independent.

3.3.3 Conclusion for GLMM

Overall, the logistic GLMM indicates that, after accounting for subject-specific baseline risk via a random intercept, there is no clear evidence that education level, time (visit), or their interaction meaningfully predicts the probability of poor cognition. The very high ICC and large random-effect variance show that most variability in poor cognition lies between individuals rather than within individuals over time, and the diagnostics point to quasi-complete separation, with many subjects never experiencing poor cognition. Correspondingly, the ROC curve shows only modest discrimination ($AUC \approx 0.67$), suggesting limited predictive value for the fixed effects. Taken together, these findings imply that, in this dataset, poor cognition is largely determined by stable subject-level factors not well captured by the measured predictors, and GLMM results for RQ1 should be interpreted cautiously.

4 Discussion

Across the GLM, GEE, and GLMM frameworks, our findings collectively characterize how demographic factors and diagnostic group membership relate to cognitive status and cognitive decline in the OASIS-2 cohort. Although each model addresses a distinct research question and operates under different assumptions, the results converge to provide a coherent picture of baseline cognitive differences and longitudinal trajectories.

4.1 Research Question 1: Predictors of Poor Cognition at Final Visit (Binary Outcome)

The baseline logistic GLM identified education as the only statistically significant protective factor for poor cognition. Each additional year of education was associated with an approximately 25% reduction in the odds of impairment, whereas age and sex were not significant predictors. Model diagnostics, including the calibration plot and an AUC of about 0.71, indicated that the GLM performed reasonably well in distinguishing individuals with and without poor cognition. In the GEE framework, which provides population-average estimates, higher education emerged as the only robust predictor of poor cognition. Individuals with high education had consistently lower predicted probabilities of $MMSE < 24$ across all visits, supporting the notion that education confers a stable “cognitive reserve”-like protection rather than a time-limited advantage. Medium education, in contrast, did not differ meaningfully from low education, and neither age nor sex showed important associations with poor cognition. The non-significant Visit and Visit $\times$ Education terms suggest that, over the relatively short follow-up and with few incident cases, the overall risk of poor cognition does not systematically worsen and the gaps between education groups remain roughly constant. Together, these population-average findings indicate that educational attainment is the main driver of between-person differences in cognitive risk in this cohort, while longitudinal changes in risk are modest and difficult to detect with the available data. But the logistic GLMM did not yield meaningful fixed-effect associations. None of the predictors (age, education, sex) reached significance. The extremely high ICC of 0.985 indicated that nearly all variability in poor cognition was attributable to

between-subject differences rather than measured predictors. The modest discriminative ability further suggests limited predictive value. Collectively, these issues highlight that neither binary GEE nor GLMM is suitable for modeling poor cognition in this dataset.

4.2 Research Question 2: Predictors of Cognitive Decline (Continuous Outcome)

Shifting to the continuous MMSE outcome allowed us to avoid the separation issue while leveraging the full variability of the data.

The GLM for $\Delta MMSE$ showed no significant relationships between baseline age, education, or sex and overall change in MMSE from baseline to final visit. Multicollinearity was not a concern ($VIF \approx 1$), and diagnostics supported the adequacy of model assumptions. This suggests that demographic covariates alone are insufficient to explain inter-individual differences in cumulative MMSE change.

As for GEE, it highlights that high education level can provide stable protective effect against falling below clinically impaired ranges. And the linear mixed model (LMM) supported the GEE conclusions while offering subject-specific insight. It identified: (a) Significant decline over time; (b) High Education level as a significant protective factor and (c) Male sex as a significant risk factor. Diagnostic plots revealed mild heteroscedasticity and expected ceiling effects, but overall model fit was acceptable. Importantly, the LMM demonstrated that although individuals decline at similar rates, their baseline cognitive levels differ substantially.

4.3 Model Preference

Another important concern is the decision to dichotomize MMSE at a cutoff of 24 to define "poor cognition." While clinically intuitive, binarizing a continuous measure discards information about within-range variability and can exacerbate instability, especially when the impaired category is rare. This was evident in the logistic GLMM, where quasi-complete separation and extreme ICC values suggested that the binary outcome was largely driven by a small subset of impaired observations, leading to wide confidence intervals and imprecise odds ratios. In contrast, modeling MMSE as a continuous outcome in the linear mixed model made fuller use of the data and yielded more stable estimates of time trends and education effects, even though ceiling effects and mild deviations from normality remained. Future work could prioritize continuous or ordinal modeling of cognitive scores, reserving dichotomization for secondary, clinically focused analyses, in order to improve efficiency, reduce separation problems, and better characterize subtle trajectories of cognitive change.

Among all analytic frameworks considered, the continuous-outcome GEE and LMM models therefore provide the most reliable, interpretable, and statistically stable results. They avoid the separation problems inherent in the binary outcome, better exploit within-subject longitudinal information, and yield estimates that align with clinical expectations about gradual cognitive decline and the protective role of education. For these reasons, our primary substantive interpretations focus on the continuous GEE and LMM findings, whereas binary models are treated as secondary, complementary analyses aimed at illustrating clinically relevant impairment thresholds rather than driving inference.

4.4 Limitations

For GEE, several limitations should be acknowledged when interpreting these findings. First, the data may be subject to **informative dropout**, as participants with faster cognitive decline may be more likely to discontinue follow-up, potentially biasing longitudinal estimates even under GEE's population-average framework. Second, the MMSE measure exhibits a well-known **ceiling effect**, particularly among highly educated or cognitively intact individuals, which can compress variability at the upper end and attenuate detectable differences in decline patterns. Third, **education was categorized into three broad groups**, which simplifies interpretation but may obscure more nuanced dose-response relationships or within-group heterogeneity. Additionally, the models assume correctly specified mean structure and rely on robust standard errors to mitigate correlation misspecification, but minor deviations observed in residual diagnostics suggest that some nonlinearity or unexplained heterogeneity may remain. Lastly, the sample size within some education-visit combinations is modest, which may limit power to detect interaction effects.

What's more, poor cognition was relatively rare and many participants never became impaired, the logistic GLMM was vulnerable to quasi-complete separation, leading to unstable fixed-effect estimates, inflated standard errors, and an extremely high ICC; as a result, inferences from this model should be interpreted with particular caution. The GLM and GLMM also relied on a limited set of baseline covariates and did not incorporate potentially important time-varying factors (e.g., comorbidities, medications, lifestyle changes), raising the possibility of residual confounding. For the linear mixed model of MMSE trajectories, we treated time as linear and used a Gaussian error structure, choices that favor interpretability but may oversimplify nonlinear or floor/ceiling dynamics in cognitive change. Finally, all models were fit within a single cohort with modest sample sizes in some strata, which restricts generalizability and limits power to detect subtle effects or higher-order interactions, particularly when outcomes are skewed or clustered at the extremes.

5 Conclusion

Overall, given the rarity of impairment and the strong between-subject heterogeneity in this cohort, we ultimately recommend focusing on models that treat MMSE as a continuous outcome. In particular, the linear mixed model of MMSE trajectories provides the most stable and clinically interpretable summary of change over time. If a simpler, cross-sectional approach is preferred, a GLM on the continuous MMSE score at baseline—and, secondarily, a logistic GLM using only first-visit data for the binary impairment outcome—offers the clearest and most reliable set of effect estimates for this dataset.

6 Data Availability

OASIS-2 data could be downloaded through kaggle (see ['/Link.txt'](#) for more information). For convenience, we also upload the data we used in github repo (see ['/data/oasis_longitudinal.csv'](#))

7 Code Availability

All codes are available at ['/code'](#). The code for [.Qmd](#) reproduction is also provided. Hope you all would enjoy our project!

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